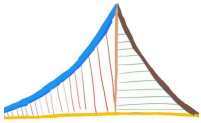
Math for EE



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About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. GATE AE 2026

A. Class-12

1.1.1 Find $|AB|$, if $A = \begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix}$.

Solution: Using the multiplicative property of determinants for square matrices:

$$|AB| = |A| \cdot |B|$$

$$|A| = \begin{vmatrix} 0 & -1 \\ 0 & 2 \end{vmatrix} = (0)(2) - (-1)(0) = 0 - 0 = 0$$

$$|B| = \begin{vmatrix} 3 & 5 \\ 0 & 0 \end{vmatrix} = (3)(0) - (5)(0) = 0 - 0 = 0$$

$$|AB| = |A| \cdot |B| = 0 \times 0 = 0$$

1.1.2 Differentiate e^{3x} with respect to x.

Solution: Given function be:

$$y = e^{3x}$$

$$\frac{d}{dx}[e^u] = e^u \frac{du}{dx}$$

Let $u = 3x$:

$$\frac{du}{dx} = \frac{d}{dx}(3x) = 3$$

Differentiating y with respect to x:

$$\frac{dy}{dx} = e^{3x} \frac{d}{dx}(3x) = e^{3x} \times 3 = 3e^{3x}$$

1.1.3 Find the order and degree (if defined) of the differential equation

$$\frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 = 2x^2 \log \left(\frac{d^2y}{dx^2} \right).$$

Solution: The given differential equation is:

$$\frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 = 2x^2 \log \left(\frac{d^2y}{dx^2} \right)$$

Order

The order of a differential equation is the order of the highest derivative appearing in it.

The derivatives appearing in the given equation are:

$$\frac{dy}{dx} \quad \text{(first derivative)} \quad \text{and} \quad \frac{d^2y}{dx^2} \quad \text{(second derivative)}$$

The highest-order derivative present is $\frac{d^2y}{dx^2}$.
Therefore:

$$\text{Order} = 2$$

Degree:
The degree of a differential equation is defined as the exponent (power) of the highest-order derivative, provided the differential equation can be expressed as a polynomial equation in terms of its derivatives.
In the given equation, the term:

$$\log\left(\frac{d^2y}{dx^2}\right)$$

contains a derivative inside a logarithmic transcendental function. Hence, the equation cannot be expressed as a polynomial in the derivatives $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.
Therefore:

$$\text{Degree is not defined.}$$

1.1.4 Find the direction cosines of a line which makes equal angles with the coordinate axes.
Solution: Let the line make angles α, β, and γ with the positive directions of the x, y, and z-axes, respectively.
The direction cosines of the line are defined as:

$$l = \cos\alpha, \quad m = \cos\beta, \quad n = \cos\gamma$$

Since the line makes equal angles with the coordinate axes[cite: 1]:

$$\alpha = \beta = \gamma \implies \cos\alpha = \cos\beta = \cos\gamma$$

which gives:

$$l = m = n$$

Using the fundamental relation between direction cosines:

$$\begin{aligned} l^2 + m^2 + n^2 &= 1 \\ l^2 + l^2 + l^2 &= 1 \\ 3l^2 &= 1 \\ l^2 &= \frac{1}{3} \\ l &= \pm\frac{1}{\sqrt{3}} \end{aligned}$$

Since $l = m = n$:

$$l = \pm\frac{1}{\sqrt{3}}, \quad m = \pm\frac{1}{\sqrt{3}}, \quad n = \pm\frac{1}{\sqrt{3}}$$

Therefore, the direction cosines of the line are:

$$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) \quad \text{or} \quad \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$$

1.1.5 A line passes through the point with position vector $2\mathbf{i}-\mathbf{j}+4\mathbf{k}$ and is in the direction of the vector $\mathbf{i}+\mathbf{j}-2\mathbf{k}$. Find the equation of the line in Cartesian form.
Solution: Let the given point through which the line passes be A, with position vector $\mathbf{a}$[cite: 3]:

$$\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 4\mathbf{k}$$

The coordinates of point A are:

$$(x_1, y_1, z_1) = (2, -1, 4)$$

The direction vector of the line is given as $\mathbf{b}$[cite: 3]:

$$\mathbf{b} = \mathbf{i} + \mathbf{j} - 2\mathbf{k}$$

Thus, the direction ratios of the line are:

$$a = 1, \quad b = 1, \quad c = -2$$

The Cartesian equation of a line passing through the point (x_1, y_1, z_1) with direction ratios a, b, c is given by:

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

Substituting the coordinates and direction ratios:

$$\frac{x - 2}{1} = \frac{y - (-1)}{1} = \frac{z - 4}{-2}$$

Simplifying the equation:

$$\frac{x - 2}{1} = \frac{y + 1}{1} = \frac{z - 4}{-2}$$

Therefore, the equation of the line in Cartesian form is:

$$\frac{x-2}{1} = \frac{y+1}{1} = \frac{4-z}{2} \quad \left(\text{or } \frac{x-2}{1} = \frac{y+1}{1} = \frac{z-4}{-2}\right)$$

1.1.6 Find

$$\int (3 - 2x - x^2)\, dx.$$

Solution: Let the given indefinite integral be:

$$I = \int (3 - 2x - x^2)\, dx$$

Using the linearity property of integration:

$$I = \int 3\, dx - \int 2x\, dx - \int x^2\, dx$$

Applying the standard power rule of integration $\int x^n\, dx = \dfrac{x^{n+1}}{n+1} + C$ (for $n \neq -1$):

$$\int 3\, dx = 3x$$
$$\int 2x\, dx = 2\left(\frac{x^2}{2}\right) = x^2$$
$$\int x^2\, dx = \frac{x^3}{3}$$

Combining the integrated terms and adding the arbitrary constant of integration C:

$$I = 3x - x^2 - \frac{x^3}{3} + C$$

Therefore:

$$\int (3 - 2x - x^2)\, dx = 3x - x^2 - \frac{1}{3}x^3 + C$$

1.1.7 If

$$A = \begin{bmatrix} p & 2 \\ 2 & p \end{bmatrix} \quad \text{and} \quad |A^3| = 125,$$

then find the values of p.

Solution: Given matrix A:

$$A = \begin{bmatrix} p & 2 \\ 2 & p \end{bmatrix}$$

Evaluating the determinant of matrix A:

$$|A| = \begin{vmatrix} p & 2 \\ 2 & p \end{vmatrix} = (p)(p) - (2)(2) = p^2 - 4$$

Using the determinant power property for square matrices, $|A^n| = |A|^n$:

$$|A^3| = |A|^3$$

Given that $|A^3| = 125$:

$$|A|^3 = 125 \implies |A|^3 = 5^3 \implies |A| = 5$$

Equating the two expressions for $|A|$:

$$p^2 - 4 = 5 \implies p^2 = 9 \implies p = \pm 3$$

Therefore, the values of p are 3 and -3.

1.1.8 Examine whether the operation $*$ defined on the set of all real numbers by

$$a * b = a^2 + b^2$$

is a binary operation or not, and if it is a binary operation, find whether it is associative or not.

Solution:

An operation $*$ on the set of real numbers $\mathbb{R}$ is a binary operation if for every pair $(a, b) \in \mathbb{R} \times \mathbb{R}$, the image $a * b \in \mathbb{R}$.

For any $a, b \in \mathbb{R}$:

$$a \in \mathbb{R} \implies a^2 \in \mathbb{R} \qquad\qquad b \in \mathbb{R} \implies b^2 \in \mathbb{R}$$

Since the sum of two real numbers is always a real number:

$$a^2 + b^2 \in \mathbb{R} \implies a * b \in \mathbb{R} \quad \forall a, b \in \mathbb{R}$$

Therefore, $*$ is a well-defined binary operation on $\mathbb{R}$.

The operation $*$ is associative on $\mathbb{R}$ if for all $a, b, c \in \mathbb{R}$:

$$(a * b) * c = a * (b * c)$$

Evaluating the Left-Hand Side (LHS):

$$(a * b) * c = (a^2 + b^2) * c = (a^2 + b^2)^2 + c^2 = a^4 + 2a^2b^2 + b^4 + c^2$$

Evaluating the Right-Hand Side (RHS):

$$a * (b * c) = a * (b^2 + c^2) = a^2 + (b^2 + c^2)^2 = a^2 + b^4 + 2b^2c^2 + c^4$$

In general:

$$(a^2 + b^2)^2 + c^2 \neq a^2 + (b^2 + c^2)^2$$

Choose $a = 1$, $b = 2$, and $c = 3 \in \mathbb{R}$:

$$(1 * 2) * 3 = (1^2 + 2^2) * 3 = (1 + 4) * 3 = 5 * 3 = 5^2 + 3^2 = 25 + 9 = 34$$

$$1 * (2 * 3) = 1 * (2^2 + 3^2) = 1 * (4 + 9) = 1 * 13 = 1^2 + 13^2 = 1 + 169 = 170$$

Since $(1 * 2) * 3 \neq 1 * (2 * 3)$ $(34 \neq 170)$, the operation $*$ is not associative.

Conclusion: The operation $*$ is a binary operation, but it is not associative.

1.1.9 If $|\mathbf{a}| = 2$, $|\mathbf{b}| = 7$ and

$$\mathbf{a} \times \mathbf{b} = 3\mathbf{i} + 2\mathbf{j} + 6\mathbf{k},$$

find the angle between $\mathbf{a}$ and $\mathbf{b}$.

Solution: Let θ be the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$, where $0 \leq \theta \leq \pi$.

Given[cite: 1]:

$$|\mathbf{a}| = 2 \qquad\qquad |\mathbf{b}| = 7$$
$$\mathbf{a} \times \mathbf{b} = 3\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$$

Evaluating the magnitude of the cross product vector $\mathbf{a} \times \mathbf{b}$:

$$|\mathbf{a} \times \mathbf{b}| = \sqrt{3^2 + 2^2 + 6^2} = \sqrt{9 + 4 + 36} = \sqrt{49} = 7$$

By the definition of the vector cross product:

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta$$

Substituting the known values:

$$7 = (2)(7)\sin\theta \implies 7 = 14\sin\theta$$
$$\sin\theta = \frac{7}{14} = \frac{1}{2}$$

Since the angle between two vectors lies in the interval $[0, \pi]$:

$$\theta = \frac{\pi}{6} \quad \text{or} \quad \theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

Therefore, the angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{\pi}{6}$ (or $30°$) or $\frac{5\pi}{6}$ (or $150°$).

1.1.10 Find the volume of a cuboid whose edges are given by

$$-3\mathbf{i} + 7\mathbf{j} + 5\mathbf{k}, -5\mathbf{i} + 7\mathbf{j} - 3\mathbf{k}, \mathbf{7i} - 5\mathbf{j} - 3\mathbf{k}.$$

Solution: The volume V of a parallelepiped with coterminous edges $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ is given by the magnitude of their scalar triple product:

$$V = |[\mathbf{a}, \mathbf{b}, \mathbf{c}]| = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$$

Expressing the scalar triple product in determinant form:

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \begin{vmatrix} -3 & 7 & 5 \\ -5 & 7 & -3 \\ 7 & -5 & -3 \end{vmatrix}$$

Expanding the determinant along the first row:

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}] = -3\begin{vmatrix} 7 & -3 \\ -5 & -3 \end{vmatrix} - 7\begin{vmatrix} -5 & -3 \\ 7 & -3 \end{vmatrix} + 5\begin{vmatrix} -5 & 7 \\ 7 & -5 \end{vmatrix}$$
$$= -3\left[(7)(-3) - (-3)(-5)\right] - 7\left[(-5)(-3) - (-3)(7)\right] + 5\left[(-5)(-5) - (7)(7)\right]$$

Since volume is a non-negative scalar quantity:

$$V = |-264| = 264 \text{ cubic units}$$

Therefore, the volume of the given solid is 264 cubic units.

1.1.11 A coin is tossed 5 times. What is the probability of getting (i) 3 heads, (ii) at most 3 heads?

Solution: Tossing a fair coin represents a sequence of Bernoulli trials where:

$$\text{Probability of getting a head (success), } p = \frac{1}{2}$$

$$\text{Probability of getting a tail (failure), } q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}$$

The probability of obtaining exactly k heads in $n = 5$ tosses is given by the binomial distribution:

$$P(X = k) = \binom{n}{k} p^k q^{n-k} = \binom{5}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{5-k} = \binom{5}{k} \left(\frac{1}{2}\right)^5 = \frac{\binom{5}{k}}{32}$$

Substituting $k = 3$:

$$P(X = 3) = \binom{5}{3} \left(\frac{1}{2}\right)^5 = \frac{5!}{3!(5-3)!} \times \frac{1}{32} = \frac{5 \times 4}{2 \times 1} \times \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$$

$$P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$$

$$P(X \leq 3) = 1 - P(X > 3) = 1 - [P(X = 4) + P(X = 5)]$$

$$P(X = 4) = \binom{5}{4} \left(\frac{1}{2}\right)^5 = \frac{5}{32}$$
$$P(X = 5) = \binom{5}{5} \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

Substituting into the complement relation:

$$P(X \leq 3) = 1 - \left(\frac{5}{32} + \frac{1}{32}\right) = 1 - \frac{6}{32} = \frac{26}{32} = \frac{13}{16}$$

The probability of getting 3 heads is $\frac{5}{16}$.
The probability of getting at most 3 heads is $\frac{13}{16}$.

1.1.12 Find the probability distribution of X, the number of heads in a simultaneous toss of two coins.

Solution: When two coins are tossed simultaneously, the sample space S is given by:

$$S = \{HH,\ HT,\ TH,\ TT\}$$

The total number of elementary events is $n(S) = 4$.

$X = 0$: No heads appear, i.e., outcome $\{TT\}$:

$$P(X = 0) = \frac{n(\{TT\})}{n(S)} = \frac{1}{4}$$

$X = 1$: Exactly one head appears, i.e., outcomes $\{HT, TH\}$:

$$P(X = 1) = \frac{n(\{HT, TH\})}{n(S)} = \frac{2}{4} = \frac{1}{2}$$

$X = 2$: Two heads appear, i.e., outcome $\{HH\}$:

$$P(X = 2) = \frac{n(\{HH\})}{n(S)} = \frac{1}{4}$$

Verifying the sum of probabilities over the state space:

$$\sum P(X) = P(X = 0) + P(X = 1) + P(X = 2) = \frac{1}{4} + \frac{2}{4} + \frac{1}{4} = 1$$

1.1.13 Find:

$$\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x}\, dx.$$

Solution:

$$\begin{aligned}\frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} &= \frac{\sin^3 x}{\sin^2 x \cos^2 x} + \frac{\cos^3 x}{\sin^2 x \cos^2 x} = \frac{\sin x}{\cos^2 x} + \frac{\cos x}{\sin^2 x} \\ &= \left(\frac{\sin x}{\cos x}\right)\left(\frac{1}{\cos x}\right) + \left(\frac{\cos x}{\sin x}\right)\left(\frac{1}{\sin x}\right) = \sec x \tan x + \csc x \cot x\end{aligned}$$

$$I = \int (\sec x \tan x + \csc x \cot x)\, dx = \int \sec x \tan x\, dx + \int \csc x \cot x\, dx$$

$$I = \sec x - \csc x + C$$

1.1.14 Find:

$$\int (x-1) e^{x^3 - x}\, dx.$$

Solution: Expressing the numerator in terms of $(x - 1)$:

$$\frac{x-3}{(x-1)^3} = \frac{(x-1)-2}{(x-1)^3} = \frac{x-1}{(x-1)^3} - \frac{2}{(x-1)^3} = \frac{1}{(x-1)^2} - \frac{2}{(x-1)^3}$$

Substituting this back into the integral:

$$I = \int e^x \left[\frac{1}{(x-1)^2} - \frac{2}{(x-1)^3}\right] dx$$

Let:

$$f(x) = \frac{1}{(x-1)^2} = (x-1)^{-2}$$

Differentiating $f(x)$ with respect to x:

$$f'(x) = -2(x-1)^{-3} = -\frac{2}{(x-1)^3}$$

The integrand is now in the standard form $\int e^x \left[f(x) + f'(x)\right] dx$:

$$I = \int \frac{e^x}{(x-1)^2}\, dx - \int \frac{2e^x}{(x-1)^3}\, dx$$

Integrating the first integral by parts:

$$\int u\, dv = uv - \int v\, du$$

Setting $u = \dfrac{1}{(x-1)^2}$ and $dv = e^x\, dx \implies du = -\dfrac{2}{(x-1)^3}\, dx$ and $v = e^x$:

$$\begin{aligned}\int \frac{e^x}{(x-1)^2}\, dx &= \frac{e^x}{(x-1)^2} - \int e^x \left(-\frac{2}{(x-1)^3}\right) dx \\ &= \frac{e^x}{(x-1)^2} + \int \frac{2e^x}{(x-1)^3}\, dx\end{aligned}$$

Substituting this expression back into I:

$$I = \left[\frac{e^x}{(x-1)^2} + \int \frac{2e^x}{(x-1)^3}\, dx\right] - \int \frac{2e^x}{(x-1)^3}\, dx + C = \frac{e^x}{(x-1)^2} + C$$

1.1.15 If $P(\text{not } A) = 0.7$, $P(B) = 0.7$ and $P(B/A) = 0.5$, then find $P(A/B)$.
Solution: First, find the probability of event A using the complement rule:

$$P(A) = 1 - P(A') = 1 - 0.7 = 0.3$$

Using the definition of conditional probability for $P(B/A)$:

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

Substituting the known values to determine the joint probability $P(A \cap B)$:

$$\begin{aligned}0.5 &= \frac{P(A \cap B)}{0.3} \\ P(A \cap B) &= 0.5 \times 0.3 = 0.15\end{aligned}$$

Now, evaluate the conditional probability $P(A/B)$[cite: 1]:

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{0.15}{0.7} = \frac{15}{70} = \frac{3}{14} \approx 0.2143$$

1.1.16 Find the general solution of the differential equation

$$\frac{dy}{dx} = e^{x+y}.$$

Solution: The given first-order differential equation is[cite: 1]:

$$\frac{dy}{dx} = e^{x+y} \implies \frac{dy}{dx} = e^x e^y$$

Separating the variables:

$$\frac{1}{e^y}\,dy = e^x\,dx \implies e^{-y}\,dy = e^x\,dx$$

$$-e^{-y} = e^x + C_1$$

where C_1 is an arbitrary constant of integration.
Rearranging the terms:

$$e^x + e^{-y} = -C_1 \implies e^x + e^{-y} = C$$

1.1.17 Find the value of

$$\sin\left(\cos^{-1}\frac{4}{5} + \tan^{-1}\frac{2}{3}\right).$$

Solution: Expressing $\cos^{-1}\left(\frac{4}{5}\right)$ in terms of $\tan^{-1}$:

$$\cos^{-1}\frac{4}{5} = \tan^{-1}\left(\frac{\sqrt{1-(4/5)^2}}{4/5}\right) = \tan^{-1}\left(\frac{3/5}{4/5}\right) = \tan^{-1}\frac{3}{4}$$

Using the addition formula $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\dfrac{x+y}{1-xy}\right)$ for $xy = \frac{3}{4}\times\frac{2}{3} = \frac{1}{2} < 1$:

$$\tan^{-1}\frac{3}{4} + \tan^{-1}\frac{2}{3} = \tan^{-1}\left(\frac{\frac{3}{4}+\frac{2}{3}}{1-\frac{3}{4}\cdot\frac{2}{3}}\right) = \tan^{-1}\left(\frac{\frac{17}{12}}{1-\frac{6}{12}}\right) = \tan^{-1}\left(\frac{17/12}{6/12}\right)$$
$$= \tan^{-1}\frac{17}{6}$$

Let $\theta = \tan^{-1}\left(\frac{17}{6}\right)$, then $\sin\theta = \dfrac{17}{\sqrt{17^2+6^2}} = \dfrac{17}{\sqrt{289+36}} = \dfrac{17}{\sqrt{325}} = \dfrac{17}{5\sqrt{13}} = \dfrac{17\sqrt{13}}{65}$.

1.1.18 Using properties of determinants, show that

$$\begin{vmatrix} 3a & -a+b & -a+c \\ -b+a & 3b & -b+c \\ -c+a & -c+b & 3c \end{vmatrix} = 3(a+b+c)(ab+bc+ca).$$

Solution:
Column Transformation to extract the factor $(a+b+c)$
Applying the elementary column operation $C_1 \to C_1 + C_2 + C_3$: Substituting back

into the determinant:

$$\Delta = \begin{vmatrix} a+b+c & b-a & c-a \\ a+b+c & 3b & c-b \\ a+b+c & b-c & 3c \end{vmatrix}$$

Factoring out the common scalar $(a+b+c)$ from column C_1:

$$\Delta = (a+b+c)\begin{vmatrix} 1 & b-a & c-a \\ 1 & 3b & c-b \\ 1 & b-c & 3c \end{vmatrix}$$

Row Transformations to generate leading zeros
Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$:
For $R_2 \to R_2 - R_1$:

$$1-1=0 \implies 3b-(b-a) = 2b+a = a+2b \implies (c-b)-(c-a) = a-b$$

For $R_3 \to R_3 - R_1$:

$$1-1=0 \implies (b-c)-(b-a) = a-c \implies 3c-(c-a) = 2c+a = a+2c$$

The determinant transforms to:

$$\Delta = (a+b+c)\begin{vmatrix} 1 & b-a & c-a \\ 0 & a+2b & a-b \\ 0 & a-c & a+2c \end{vmatrix}$$

$$\begin{aligned} \Delta &= (a+b+c)\left[1 \cdot \begin{vmatrix} a+2b & a-b \\ a-c & a+2c \end{vmatrix} - 0 + 0\right] \\ &= (a+b+c)\left[(a+2b)(a+2c) - (a-b)(a-c)\right] \end{aligned}$$

$$(a+2b)(a+2c) = a^2 + 2ac + 2ab + 4bc \implies (a-b)(a-c) = a^2 - ac - ab + bc$$

$$\begin{aligned} (a+2b)(a+2c) - (a-b)(a-c) &= (a^2 + 2ab + 2ac + 4bc) - (a^2 - ab - ac + bc) = 3ab \\ &= 3(ab+bc+ca) \end{aligned}$$

$$\Delta = 3(a+b+c)(ab+bc+ca)$$

1.1.19 Check whether the relation R defined on the set $A = \{1,2,3,4,5,6\}$ a $R = \{(a,b) : b = a+1\}$ is reflexive, symmetric or transitive.
Solution: Listing all ordered pairs $(a,b) \in R$:

- For $a=1 \implies b = 1+1 = 2 \in A \implies (1,2) \in R$
- For $a=2 \implies b = 2+1 = 3 \in A \implies (2,3) \in R$
- For $a=3 \implies b = 3+1 = 4 \in A \implies (3,4) \in R$
- For $a=4 \implies b = 4+1 = 5 \in A \implies (4,5) \in R$

- For $a = 5 \implies b = 5 + 1 = 6 \in A \implies (5, 6) \in R$
- For $a = 6 \implies b = 6 + 1 = 7 \notin A$

Thus, in roster form:

$$R = \{(1, 2),\ (2, 3),\ (3, 4),\ (4, 5),\ (5, 6)\}$$

Reflexivity: A relation R on set A is reflexive if $(a, a) \in R$ for all $a \in A$.

$$(1, 1) \notin R \quad (\text{since } 1 \neq 1 + 1)$$

Symmetry: A relation R on set A is symmetric if $(a, b) \in R \implies (b, a) \in R$.

$$(1, 2) \in R, \quad \text{but} \quad (2, 1) \notin R \quad (\text{since } 1 \neq 2 + 1)$$

Transitivity: A relation R on set A is transitive if $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$.

$$(1, 2) \in R \quad \text{and} \quad (2, 3) \in R, \quad \text{but} \quad (1, 3) \notin R \quad (\text{since } 3 \neq 1 + 1)$$

Conclusion: The relation R is neither reflexive, nor symmetric, nor transitive.

1.1.20 Let $f : \mathbb{N} \to Y$ be a function defined as $f(x) = 4x + 3$, where

$$Y = \{y \in \mathbb{N} : y = 4x + 3 \text{ for some } x \in \mathbb{N}\}.$$

Show that f is invertible. Find its inverse.

Solution: A function is invertible if and only if it is a bijection (both one-one and onto).

Injectivity (One-One): Let $x_1, x_2 \in \mathbb{N}$ such that:

$$f(x_1) = f(x_2)$$
$$4x_1 + 3 = 4x_2 + 3 \implies 4x_1 = 4x_2 \implies x_1 = x_2$$

Since $f(x_1) = f(x_2) \implies x_1 = x_2$ for all $x_1, x_2 \in \mathbb{N}$, the function f is injective (one-one).

Surjectivity (Onto): Let $y \in Y$ be an arbitrary element. By the definition of set Y[cite: 1]:

$$y = 4x + 3 \quad \text{for some } x \in \mathbb{N}$$

Solving for x in terms of y:

$$4x = y - 3 \implies x = \frac{y - 3}{4}$$

Since $y \in Y$, by construction $\dfrac{y - 3}{4}$ is guaranteed to be a natural number $x \in \mathbb{N}$ such that:

$$f(x) = f\left(\frac{y - 3}{4}\right) = 4\left(\frac{y - 3}{4}\right) + 3 = (y - 3) + 3 = y$$

Thus, every element in codomain Y has a pre-image in domain $\mathbb{N}$. Therefore, f is surjective (onto).

Finding the Inverse Function f^{-1}: Define a function $g : Y \to \mathbb{N}$ by:

$$g(y) = \frac{y-3}{4}$$

$$f^{-1}(y) = \frac{y-3}{4}$$

1.1.21 If $x\sqrt{1+y} + y\sqrt{1+x} = 0, \qquad x \neq y$,prove that $\frac{dy}{dx} = -\frac{1}{(x+1)^2}$.
Solution:

$$x^2(1+y) = y^2(1+x) \implies x^2 + x^2y = y^2 + xy^2 \implies (x^2 - y^2) + (x^2y - xy^2) = 0$$

Factoring the algebraic expression:

$$(x-y)(x+y) + xy(x-y) = 0 \implies (x-y)(x+y+xy) = 0$$

$$x + y + xy = 0$$

$$y(1+x) = -x \implies y = -\frac{x}{1+x}$$

Rewriting the expression to simplify differentiation:

$$y = -\frac{(x+1)-1}{x+1} = -\left(1 - \frac{1}{x+1}\right) = -1 + (x+1)^{-1}$$

Differentiating both sides with respect to x:

$$\frac{dy}{dx} = \frac{d}{dx}\left[-1 + (x+1)^{-1}\right] = -\frac{1}{(x+1)^2}$$

1.1.22 If $(\cos x)^y = (\sin y)^x$, find $\dfrac{dy}{dx}$.
Solution: Taking the natural logarithm on both sides:

$$\ln\left((\cos x)^y\right) = \ln\left((\sin y)^x\right)$$
$$y\ln(\cos x) = x\ln(\sin y)$$

Differentiating both sides with respect to x:

$$\frac{d}{dx}\left[y\ln(\cos x)\right] = \frac{d}{dx}\left[x\ln(\sin y)\right]$$
$$\frac{dy}{dx}\cdot\ln(\cos x) + y\cdot\frac{d}{dx}[\ln(\cos x)] = \frac{dx}{dx}\cdot\ln(\sin y) + x\cdot\frac{d}{dx}[\ln(\sin y)]$$
$$\frac{dy}{dx}\ln(\cos x) + y\left(\frac{1}{\cos x}\cdot(-\sin x)\right) = 1\cdot\ln(\sin y) + x\left(\frac{1}{\sin y}\cdot\cos y\cdot\frac{dy}{dx}\right)$$
$$\frac{dy}{dx}\ln(\cos x) - y\tan x = \ln(\sin y) + x\cot y\frac{dy}{dx}$$

Collecting all terms involving $\dfrac{dy}{dx}$ on the left-hand side:

$$\frac{dy}{dx}\ln(\cos x) - x\cot y\frac{dy}{dx} = \ln(\sin y) + y\tan x$$
$$\frac{dy}{dx}\left[\ln(\cos x) - x\cot y\right] = \ln(\sin y) + y\tan x$$

Solving for $\dfrac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{\ln(\sin y) + y\tan x}{\ln(\cos x) - x\cot y}$$

1.1.23 Prove that $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$, and hence evaluate $\int_0^{\pi/2} \frac{x}{\sin x+\cos x}\,dx$.
Solution: When $x = 0$, $t = a$, and when $x = a$, $t = 0$.

$$I_0 = \int_a^0 f(a-t)(-dt) = -\int_a^0 f(a-t)\,dt = \int_0^a f(a-t)\,dt$$

Changing the dummy variable from t to x:

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

Evaluation of the integral:

$$I = \int_0^{\pi/2} \frac{x}{\sin x + \cos x}\,dx$$

Applying the property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$:

$$I = \int_0^{\pi/2} \frac{\frac{\pi}{2} - x}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)}\,dx = \int_0^{\pi/2} \frac{\frac{\pi}{2} - x}{\cos x + \sin x}\,dx$$

Adding the two expressions for I:

$$2I = \int_0^{\pi/2} \frac{x + \left(\frac{\pi}{2} - x\right)}{\sin x + \cos x}\,dx \implies 2I = \frac{\pi}{2}\int_0^{\pi/2} \frac{1}{\sin x + \cos x}\,dx \implies I = \frac{\pi}{4}\int_0^{\pi/2} \frac{}{\sin x}$$

Transforming the denominator:

$$\sin x + \cos x = \sqrt{2}\left(\frac{1}{\sqrt{2}}\cos x + \frac{1}{\sqrt{2}}\sin x\right) = \sqrt{2}\cos\left(x - \frac{\pi}{4}\right)$$

Substituting back:

$$I = \frac{\pi}{4\sqrt{2}}\int_0^{\pi/2} \sec\left(x - \frac{\pi}{4}\right)\,dx$$

Substitute $u = x - \frac{\pi}{4} \implies du = dx$. When $x = 0$, $u = -\frac{\pi}{4}$; when $x = \frac{\pi}{2}$, $u = \frac{\pi}{4}$:

$$I = \frac{\pi}{4\sqrt{2}}\int_{-\pi/4}^{\pi/4} \sec u\,du = \frac{\pi}{4\sqrt{2}} \cdot 2\int_0^{\pi/4} \sec u\,du = \frac{\pi}{2\sqrt{2}}\ln(\sqrt{2}+1)$$

1.1.24 Find:

$$\int \frac{x^2+x+1}{(x+2)(x^2+1)}\,dx.$$

Solution:

$$\frac{x^2+x+1}{(x+2)(x^2+1)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+1}$$

Expanding and collecting coefficients of like powers of x:

$$x^2+x+1 = A(x^2+1) + B(x^2+2x) + C(x+2) \implies x^2+x+1 = (A+B)x^2 + (2B$$

Substituting A, B, and C back into the partial fraction decomposition:

$$\frac{x^2+x+1}{(x+2)(x^2+1)} = \frac{3/5}{x+2} + \frac{\frac{2}{5}x+\frac{1}{5}}{x^2+1} = \frac{3}{5}\left(\frac{1}{x+2}\right) + \frac{1}{5}\left(\frac{2x}{x^2+1}\right) + \frac{1}{5}\left(\frac{1}{x^2+1}\right)$$

Integrating term-by-term:

$$I = \frac{3}{5}\int \frac{1}{x+2}\,dx + \frac{1}{5}\int \frac{2x}{x^2+1}\,dx + \frac{1}{5}\int \frac{1}{x^2+1}\,dx$$

Using standard integration identities:

$$\int \frac{1}{x+2}\,dx = \ln|x+2| \implies \int \frac{2x}{x^2+1}\,dx = \ln(x^2+1) \quad (\text{since } x^2+1>0) \implies \int \frac{}{x}$$

Combining the evaluated terms:

$$I = \frac{3}{5}\ln|x+2| + \frac{1}{5}\ln(x^2+1) + \frac{1}{5}\tan^{-1}x + C$$

1.1.25 Solve the differential equation

$$x\frac{dy}{dx} = y - x\tan\left(\frac{y}{x}\right).$$

Solution: Dividing both sides by x (for $x \neq 0$):

$$\frac{dy}{dx} = \frac{y}{x} - \tan\left(\frac{y}{x}\right)$$

Substituting $y = vx$ and $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$ into the differential equation:

$$v + x\frac{dv}{dx} = v - \tan v \implies x\frac{dv}{dx} = -\tan v$$

Rearranging the variables v and x:

$$\frac{dv}{\tan v} = -\frac{dx}{x} \implies \cot v\,dv = -\frac{dx}{x}$$

Integrating both sides:

$$\int \cot v \, dv = -\int \frac{1}{x} \, dx$$
$$\ln|\sin v| = -\ln|x| + \ln C_1$$

where $\ln C_1$ is an arbitrary constant ($C_1 > 0$).
Using logarithmic properties:

$$\ln|\sin v| + \ln|x| = \ln C_1 \implies x \sin v = C$$

where $C = \pm C_1$ is an arbitrary real non-zero constant.
Substituting $v = \dfrac{y}{x}$ back into the solution:

$$x \sin\left(\frac{y}{x}\right) = C$$

1.1.26 Solve the differential equation

$$\frac{dy}{dx} = -\frac{x + y\cos x}{1 + \sin x}.$$

Solution:

$$\frac{dy}{dx} = -\frac{x}{1+\sin x} - \frac{y\cos x}{1+\sin x}$$
$$\frac{dy}{dx} + \left(\frac{\cos x}{1+\sin x}\right) y = -\frac{x}{1+\sin x}$$

This is a linear first-order differential equation of the standard form:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where:

$$P(x) = \frac{\cos x}{1+\sin x}, \qquad Q(x) = -\frac{x}{1+\sin x}$$

Integrating factor:

$$\text{I.F.} = e^{\int P(x)\,dx} = e^{\int \frac{\cos x}{1+\sin x}\,dx}$$

Let $u = 1 + \sin x \implies du = \cos x \, dx$:

$$\int \frac{\cos x}{1+\sin x} \, dx = \ln|1 + \sin x|$$

Since $1 + \sin x \geq 0$ for all real x:

$$\text{I.F.} = e^{\ln(1+\sin x)} = 1 + \sin x$$

General solution: The general solution of a linear differential equation is given by:

$$y \cdot (\text{I.F.}) = \int Q(x) \cdot (\text{I.F.}) \, dx + C$$

Substituting the values of I.F. and $Q(x)$:

$$y(1+\sin x) = \int \left(-\frac{x}{1+\sin x}\right)(1+\sin x)\,dx + C \implies y(1+\sin x) = \int (-x)\,dx + C$$

Multiplying throughout by 2:

$$2y(1+\sin x) = -x^2 + 2C \implies 2y(1+\sin x) + x^2 = C'$$

1.1.27 The scalar product of the vector $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ with a unit vector along the sum of the vectors $\mathbf{b} = 2\mathbf{i} + 4\mathbf{j} - 5\mathbf{k}, \mathbf{c} = \lambda\hat{i} + 2\mathbf{j} + 3\mathbf{k}$ is equal to 1. Find the value of λ and hence find the unit vector along $\mathbf{b} + \mathbf{c}$.

Solution:

$$\begin{aligned}
\mathbf{b} + \mathbf{c} &= (2\mathbf{i} + 4\mathbf{j} - 5\mathbf{k}) + (\lambda\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) \\
&= (2+\lambda)\mathbf{i} + (4+2)\mathbf{j} + (-5+3)\mathbf{k} \\
&= (\lambda+2)\mathbf{i} + 6\mathbf{j} - 2\mathbf{k} \\
|\mathbf{b} + \mathbf{c}| &= \sqrt{(\lambda+2)^2 + 6^2 + (-2)^2} \\
&= \sqrt{(\lambda+2)^2 + 36 + 4} \\
&= \sqrt{(\lambda+2)^2 + 40}
\end{aligned}$$

Unit vector along the sum: Let $\mathbf{u}$ denote the unit vector in the direction of $\mathbf{b} + \mathbf{c}$:

$$\mathbf{u} = \frac{\mathbf{b} + \mathbf{c}}{|\mathbf{b} + \mathbf{c}|} = \frac{(\lambda+2)\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}}{\sqrt{(\lambda+2)^2 + 40}}$$

Scalar product condition: The scalar product of $\mathbf{a}$ with the unit vector $\mathbf{u}$ is given to be 1:

$$\mathbf{a} \cdot \mathbf{u} = 1 \implies \mathbf{a} \cdot \frac{\mathbf{b} + \mathbf{c}}{|\mathbf{b} + \mathbf{c}|} = 1$$

Evaluating the dot product in the numerator:

$$\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = (\mathbf{i} + \mathbf{j} + \mathbf{k}) \cdot [(\lambda+2)\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}] = \lambda + 6$$

Substituting into the equation:

$$\frac{\lambda + 6}{\sqrt{(\lambda+2)^2 + 40}} = 1 \implies \lambda + 6 = \sqrt{(\lambda+2)^2 + 40}$$

Squaring both sides (with $\lambda + 6 > 0$):

$$\begin{aligned}
(\lambda+6)^2 &= (\lambda+2)^2 + 40 \\
\lambda^2 + 12\lambda + 36 &= (\lambda^2 + 4\lambda + 4) + 40 \\
8\lambda &= 8 \implies \lambda = 1
\end{aligned}$$

Substitute $\lambda = 1$ back into $\mathbf{b} + \mathbf{c}$:

$$\mathbf{b} + \mathbf{c} = (1+2)\mathbf{i} + 6\mathbf{j} - 2\mathbf{k} = 3\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}$$
$$|\mathbf{b} + \mathbf{c}| = \sqrt{3^2 + 6^2 + (-2)^2} = \sqrt{9 + 36 + 4} = \sqrt{49} = 7$$

Therefore, the unit vector along $\mathbf{b} + \mathbf{c}$ is:

$$\mathbf{u} = \frac{3\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}}{7} = \frac{3}{7}\mathbf{i} + \frac{6}{7}\mathbf{j} - \frac{2}{7}\mathbf{k}$$

1.1.28 If $(a + bx)e^{y/x} = x$,then prove that $x^3 \frac{d^2y}{dx^2} = \left(x\frac{dy}{dx} - y\right)^2$.

Solution: Taking the natural logarithm on both sides:

$$\ln\left((a + bx)e^{y/x}\right) = \ln x$$
$$\ln(a + bx) + \ln\left(e^{y/x}\right) = \ln x$$
$$\ln(a + bx) + \frac{y}{x} = \ln x$$
$$\frac{y}{x} = \ln x - \ln(a + bx)$$

Differentiating both sides with respect to x:

$$\frac{d}{dx}\left(\frac{y}{x}\right) = \frac{d}{dx}\left[\ln x - \ln(a + bx)\right]$$

Using the quotient rule on the left-hand side and standard logarithmic differentiation on the right-hand side:

$$\frac{x\frac{dy}{dx} - y(1)}{x^2} = \frac{1}{x} - \frac{b}{a + bx}$$
$$\frac{x\frac{dy}{dx} - y}{x^2} = \frac{(a + bx) - bx}{x(a + bx)}$$
$$\frac{x\frac{dy}{dx} - y}{x^2} = \frac{a}{x(a + bx)}$$

Multiplying both sides by x^2:

$$x\frac{dy}{dx} - y = \frac{ax}{a + bx} \tag{1.1.28.1}$$

Differentiating both sides of equation (1.1.28.1) with respect to x:

$$\frac{d}{dx}\left(x\frac{dy}{dx}-y\right)=\frac{d}{dx}\left(\frac{ax}{a+bx}\right)$$

$$\left(1\cdot\frac{dy}{dx}+x\frac{d^2y}{dx^2}\right)-\frac{dy}{dx}=\frac{(a+bx)\frac{d}{dx}(ax)-(ax)\frac{d}{dx}(a+bx)}{(a+bx)^2}$$

$$x\frac{d^2y}{dx^2}=\frac{a^2+abx-abx}{(a+bx)^2}$$

$$x\frac{d^2y}{dx^2}=\frac{a^2}{(a+bx)^2}$$

$$x^3\frac{d^2y}{dx^2}=\frac{a^2x^2}{(a+bx)^2}$$

$$x^3\frac{d^2y}{dx^2}=\left(\frac{ax}{a+bx}\right)^2$$

Substituting $\dfrac{ax}{a+bx}=x\dfrac{dy}{dx}-y$ from equation (1.1.28.1):

$$x^3\frac{d^2y}{dx^2}=\left(x\frac{dy}{dx}-y\right)^2$$

1.1.29 The volume of a cube is increasing at the rate of 8 cm^3/s. How fast is the surface area increasing when the length of its edge is 12 cm?

Solution: Let the edge length of the cube at any time t be x cm.

Volume of the cube: $V=x^3$

Surface area of the cube: $S=6x^2$

Differentiating V with respect to time t:

$$\frac{dV}{dt}=\frac{d}{dt}(x^3)=3x^2\frac{dx}{dt}$$

Given that $\dfrac{dV}{dt}=8$ cm^3/s:

$$8=3x^2\frac{dx}{dt}\implies\frac{dx}{dt}=\frac{8}{3x^2}$$

Differentiating S with respect to time t:

$$\frac{dS}{dt}=\frac{d}{dt}(6x^2)=12x\frac{dx}{dt}$$

Substituting the expression for $\dfrac{dx}{dt}$:

$$\frac{dS}{dt}=12x\left(\frac{8}{3x^2}\right)=\frac{96x}{3x^2}=\frac{32}{x}$$

Evaluating at $x = 12$ cm:

$$\left.\frac{dS}{dt}\right|_{x=12} = \frac{32}{12} = \frac{8}{3} \text{ cm}^2/\text{s}$$

1.1.30 Find the Cartesian and vector equations of the plane passing through the points $A(2, 5, -3), \quad B(-2, -3, 5), \quad C(5, 3, -3)$.

Solution: Using the three-point determinant form:

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

$$\begin{vmatrix} x - 2 & y - 5 & z + 3 \\ -4 & -8 & 8 \\ 3 & -2 & 0 \end{vmatrix} = 0$$

Expanding along the first row:

$$\begin{aligned} (x-2)(16) - (y-5)(-24) + (z+3)(32) &= 0 \\ 16(x-2) + 24(y-5) + 32(z+3) &= 0 \end{aligned}$$

Dividing the entire equation by 8:

$$\begin{aligned} 2(x-2) + 3(y-5) + 4(z+3) &= 0 \\ 2x - 4 + 3y - 15 + 4z + 12 &= 0 \\ 2x + 3y + 4z - 7 = 0 \implies 2x + 3y + 4z &= 7 \end{aligned}$$

Cartesian equation: $2x + 3y + 4z = 7$ Vector equation: $\mathbf{r} \cdot (2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}) = 7$

1.1.31 Find the point on the curve $y^2 = 4x$, which is nearest to the point $(2, -8)$.

Solution:

Thus, the coordinates of P can be represented in terms of y as:

$$P\left(\frac{y^2}{4}, y\right)$$

Let the given fixed point be $A(2, -8)$.

The Euclidean distance D between $P\left(\frac{y^2}{4}, y\right)$ and $A(2, -8)$ is:

$$D = \sqrt{\left(\frac{y^2}{4} - 2\right)^2 + (y - (-8))^2} = \sqrt{\left(\frac{y^2}{4} - 2\right)^2 + (y + 8)^2}$$

Minimizing D is equivalent to minimizing the squared distance functional $Z = D^2$:

$$Z = \left(\frac{y^2}{4} - 2\right)^2 + (y + 8)^2$$

Differentiating Z with respect to y:

$$\begin{aligned}\frac{dZ}{dy} &= 2\left(\frac{y^2}{4}-2\right)\frac{d}{dy}\left(\frac{y^2}{4}-2\right)+2(y+8)\frac{d}{dy}(y+8)\\ &= 2\left(\frac{y^2}{4}-2\right)\left(\frac{2y}{4}\right)+2(y+8)(1) = y\left(\frac{y^2}{4}-2\right)+2y+16\\ &= \frac{y^3}{4}-2y+2y+16 = \frac{y^3}{4}+16\end{aligned}$$

Setting $\dfrac{dZ}{dy}=0$ for stationary points:

$$\frac{y^3}{4}+16=0 \implies \frac{y^3}{4}=-16 \implies y=-4$$

Second derivative test: Differentiating $\dfrac{dZ}{dy}$ with respect to y:

$$\frac{d^2Z}{dy^2}=\frac{d}{dy}\left(\frac{y^3}{4}+16\right)=\frac{3y^2}{4}$$

Evaluating at the critical point $y=-4$:

$$\left.\frac{d^2Z}{dy^2}\right|_{y=-4}=\frac{3(-4)^2}{4}=\frac{3(16)}{4}=12>0$$

Since $\dfrac{d^2Z}{dy^2}>0$, Z attains a local and global minimum at $y=-4$.

Substituting $y=-4$ into the relation $x=\dfrac{y^2}{4}$:

$$x=\frac{(-4)^2}{4}=\frac{16}{4}=4$$

Therefore, the point on the curve nearest to $(2,-8)$ is $(4,-4)$.

1.1.32 Find $\int_1^3 (x^2+2+e^{2x})\,dx$ as the limit of sums.

Solution: By definition, the definite integral as the limit of sums is given by:

$$\int_a^b f(x)\,dx=\lim_{h\to 0} h\sum_{r=1}^{n} f(a+rh)$$

where $h=\dfrac{b-a}{n} \implies nh=b-a$.

Here[cite: 1]:

$$a=1, \qquad b=3 \implies nh=3-1=2$$

and the integrand is:

$$f(x)=x^2+2+e^{2x}$$

Evaluating $f(a+rh)$:

$$f(1+rh) = (1+rh)^2 + 2 + e^{2(1+rh)} = 3 + 2rh + r^2h^2 + e^2(e^{2h})^r$$

Setting up the Riemann sum:

$$S_n = \sum_{r=1}^{n} f(1+rh) = \sum_{r=1}^{n} 3 + 2h\sum_{r=1}^{n} r + h^2\sum_{r=1}^{n} r^2 + e^2\sum_{r=1}^{n}(e^{2h})^r$$

Using standard summation formulas:

$$\sum_{r=1}^{n} 3 = 3n \qquad \sum_{r=1}^{n} r = \frac{n(n+1)}{2} \qquad \sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$$

For the geometric progression with first term e^{2h} and common ratio e^{2h}:

$$\sum_{r=1}^{n}(e^{2h})^r = e^{2h}\left[\frac{(e^{2h})^n - 1}{e^{2h} - 1}\right] = e^{2h}\left[\frac{e^{2nh} - 1}{e^{2h} - 1}\right]$$

Multiplying the sum by h:

$$\begin{aligned} hS_n &= 3nh + 2h^2\left[\frac{n(n+1)}{2}\right] + h^3\left[\frac{n(n+1)(2n+1)}{6}\right] + he^2 \cdot e^{2h}\left[\frac{e^{2nh} - 1}{e^{2h} - 1}\right] \\ &= 3(nh) + (nh)(nh+h) + \frac{(nh)(nh+h)(2nh+h)}{6} + e^2(e^{2nh} - 1) \cdot e^{2h}\left(\frac{h}{e^{2h} - 1}\right) \end{aligned}$$

Substituting $nh = 2$:

$$\begin{aligned} hS_n &= 3(2) + 2(2+h) + \frac{2(2+h)(4+h)}{6} + e^2(e^4 - 1) \cdot e^{2h}\left(\frac{h}{e^{2h} - 1}\right) \\ &= 6 + 2(2+h) + \frac{(2+h)(4+h)}{3} + (e^6 - e^2) \cdot e^{2h}\left(\frac{1}{2} \cdot \frac{2h}{e^{2h} - 1}\right) \end{aligned}$$

Taking the limit as $n \to \infty$ ($h \to 0$):
Using the standard limit $\lim\limits_{t\to 0} \dfrac{e^t - 1}{t} = 1 \implies \lim\limits_{h\to 0} \dfrac{2h}{e^{2h} - 1} = 1$:

$$\begin{aligned} \int_1^3 (x^2 + 2 + e^{2x})\,dx &= \lim_{h\to 0}(hS_n) \\ &= 6 + 2(2+0) + \frac{(2+0)(4+0)}{3} + (e^6 - e^2) \cdot e^0\left(\frac{1}{2} \cdot 1\right) \\ &= 6 + 4 + \frac{8}{3} + \frac{e^6 - e^2}{2} = \frac{38}{3} + \frac{e^6 - e^2}{2} \end{aligned}$$

Therefore:

$$\int_1^3 (x^2 + 2 + e^{2x})\,dx = \frac{38}{3} + \frac{e^6 - e^2}{2}$$

1.1.33 Using integration, find the area of the triangular region whose sides have the equations $y = 2x + 1, \qquad y = 3x + 1, \qquad x = 4.$

Solution:

$$y = 2x + 1 \tag{1.1.33.1}$$
$$y = 3x + 1 \tag{1.1.33.2}$$
$$x = 4 \tag{1.1.33.3}$$

Intersection of lines (1.1.33.1) and (1.1.33.2):

$$2x + 1 = 3x + 1 \implies x = 0$$

Substituting $x = 0$ into equation (1.1.33.1):

$$y = 2(0) + 1 = 1$$

Thus, the vertex is $A(0, 1)$.
Intersection of lines (1.1.33.1) and (1.1.33.3): Substituting $x = 4$ into equation (1.1.33.1):

$$y = 2(4) + 1 = 9$$

Thus, the vertex is $B(4, 9)$.
Intersection of lines (1.1.33.2) and (1.1.33.3): Substituting $x = 4$ into equation (1.1.33.2):

$$y = 3(4) + 1 = 13$$

Thus, the vertex is $C(4, 13)$.
In the interval $x \in [0, 4]$, the line $y = 3x + 1$ lies above the line $y = 2x + 1$ since $3x + 1 \geq 2x + 1$ for all $x \geq 0$.
The area A of the region bounded between $x = 0$ and $x = 4$ is:

$$\text{Area} = \int_0^4 (y_{\text{upper}} - y_{\text{lower}})\, dx = \int_0^4 [(3x + 1) - (2x + 1)]\, dx$$
$$= \int_0^4 (3x + 1 - 2x - 1)\, dx = \int_0^4 x\, dx$$

Evaluating the definite integral:

$$\text{Area} = \left[\frac{x^2}{2}\right]_0^4 = \frac{4^2}{2} - 0 = \frac{16}{2} = 8 \text{ square units}$$

Therefore, the area of the triangular region is 8 square units.

1.1.34 If

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 1 & 2 \\ 5 & 1 & 1 \end{bmatrix},$$

find A^{-1}. Hence solve the system of equations

$$\begin{aligned} x + 3y + 4z &= 8, \\ 2x + y + 2z &= 5, \\ 5x + y + z &= 7. \end{aligned}$$

Solution: Determinant of matrix A: Expanding $|A|$ along the first row:

$$|A| = 1\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} - 3\begin{vmatrix} 2 & 2 \\ 5 & 1 \end{vmatrix} + 4\begin{vmatrix} 2 & 1 \\ 5 & 1 \end{vmatrix} = 1(-1) - 3(-8) + 4(-3) = 11$$

Let C_{ij} denote the cofactor of entry a_{ij}:

$$C_{11} = +\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = 1 - 2 = -1$$

$$C_{12} = -\begin{vmatrix} 2 & 2 \\ 5 & 1 \end{vmatrix} = -(2 - 10) = 8$$

$$C_{13} = +\begin{vmatrix} 2 & 1 \\ 5 & 1 \end{vmatrix} = 2 - 5 = -3$$

$$C_{21} = -\begin{vmatrix} 3 & 4 \\ 1 & 1 \end{vmatrix} = -(3 - 4) = 1$$

$$C_{22} = +\begin{vmatrix} 1 & 4 \\ 5 & 1 \end{vmatrix} = 1 - 20 = -19$$

$$C_{23} = -\begin{vmatrix} 1 & 3 \\ 5 & 1 \end{vmatrix} = -(1 - 15) = 14$$

$$C_{31} = +\begin{vmatrix} 3 & 4 \\ 1 & 2 \end{vmatrix} = 6 - 4 = 2$$

$$C_{32} = -\begin{vmatrix} 1 & 4 \\ 2 & 2 \end{vmatrix} = -(2 - 8) = 6$$

$$C_{33} = +\begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} = 1 - 6 = -5$$

Adjugate and inverse: The matrix of cofactors is:

$$C = \begin{bmatrix} -1 & 8 & -3 \\ 1 & -19 & 14 \\ 2 & 6 & -5 \end{bmatrix}$$

The adjugate of A is the transpose of the cofactor matrix:

$$\text{adj}(A) = C^T = \begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix}$$

Thus, the inverse matrix A^{-1} is:

$$A^{-1} = \frac{1}{|A|}\,\mathrm{adj}(A) = \frac{1}{11}\begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix}$$

Solution of the linear system: The given system of linear equations is[cite: 1]:

$$\begin{aligned} x + 3y + 4z &= 8 \\ 2x + y + 2z &= 5 \\ 5x + y + z &= 7 \end{aligned}$$

Expressing the system in canonical matrix form $AX = B$:

$$\begin{bmatrix} 1 & 3 & 4 \\ 2 & 1 & 2 \\ 5 & 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \\ 7 \end{bmatrix}$$

where:

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \qquad B = \begin{bmatrix} 8 \\ 5 \\ 7 \end{bmatrix}$$

Multiplying both sides on the left by A^{-1}:

$$X = A^{-1}B$$

Substituting A^{-1} and B:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{11}\begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix}\begin{bmatrix} 8 \\ 5 \\ 7 \end{bmatrix}$$

Evaluating the matrix multiplication:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{11}\begin{bmatrix} (-1)(8) + (1)(5) + (2)(7) \\ (8)(8) + (-19)(5) + (6)(7) \\ (-3)(8) + (14)(5) + (-5)(7) \end{bmatrix} \\ &= \frac{1}{11}\begin{bmatrix} -8 + 5 + 14 \\ 64 - 95 + 42 \\ -24 + 70 - 35 \end{bmatrix} \\ &= \frac{1}{11}\begin{bmatrix} 11 \\ 11 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

Equating corresponding entries:

$$x = 1, \qquad y = 1, \qquad z = 1$$

1.1.35 Find the inverse of the following matrix, using elementary transformations:

$$A = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}.$$

Solution: To find A^{-1} using elementary row transformations, write:

$$A = IA$$

$$\begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Elementary row operations:
Applying $R_2 \to R_2 - 2R_1$:

$$\begin{bmatrix} 2 & 0 & -1 \\ 5-2(2) & 1-2(0) & 0-2(-1) \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0-2(1) & 1-2(0) & 0-2(0) \\ 0 & 0 & 1 \end{bmatrix} A$$

$$\begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Interchanging rows $R_1 \leftrightarrow R_2$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 0 & -1 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Applying $R_2 \to R_2 - 2R_1$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 2-2(1) & 0-2(1) & -1-2(2) \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1-2(-2) & 0-2(1) & 0-2(0) \\ 0 & 0 & 1 \end{bmatrix} A$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & -5 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 5 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Interchanging rows $R_2 \leftrightarrow R_3$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & -2 & -5 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 0 & 0 & 1 \\ 5 & -2 & 0 \end{bmatrix} A$$

Applying $R_1 \to R_1 - R_2$ and $R_3 \to R_3 + 2R_2$:

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2-0 & 1-0 & 0-1 \\ 0 & 0 & 1 \\ 5+2(0) & -2+2(0) & 0+2(1) \end{bmatrix} A$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & -1 \\ 0 & 0 & 1 \\ 5 & -2 & 2 \end{bmatrix} A$$

Applying $R_1 \to R_1 + R_3$ and $R_2 \to R_2 - 3R_3$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2+5 & 1+(-2) & -1+2 \\ 0-3(5) & 0-3(-2) & 1-3(2) \\ 5 & -2 & 2 \end{bmatrix} A$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix} A$$

Since $I = A^{-1}A$, the inverse of matrix A is:

$$A^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$$

1.1.36 A company produces two types of goods, A and B, that require gold and silver. Each unit of type A requires 3 g of silver and 1 g of gold while that of type B requires 1 g of silver and 2 g of gold. The company can use at the most 9 g of silver and 8 g of gold. If each unit of type A brings a profit of Rs. 40 and that of type B Rs. 50, find the number of units of each type that the company should produce to maximize profit. Formulate the above LPP and solve it graphically and also find the maximum profit.

Solution: Let x be the number of units of type A produced. Let y be the number of units of type B produced.

The profit generated from type A is Rs. 40 per unit and from type B is Rs. 50 per unit[cite: 1]. The total profit Z to be maximized is:

$$Z = 40x + 50y$$

Silver constraint: Type A requires 3 g and type B requires 1 g of silver, with a maximum availability of 9 g[cite: 1]:

$$3x + y \leq 9$$

Gold constraint: Type A requires 1 g and type B requires 2 g of gold, with a maximum availability of 8 g[cite: 1]:

$$x + 2y \leq 8$$

Non-negativity constraints: The production quantities cannot be negative:

$$x \geq 0, \qquad y \geq 0$$

Thus, the mathematical formulation of the LPP is:

$$\begin{aligned} \text{Maximize } Z &= 40x + 50y \\ \text{subject to } 3x + y &\leq 9 \\ x + 2y &\leq 8 \\ x \geq 0, &\quad y \geq 0 \end{aligned}$$

Plotting the boundary lines associated with each constraint:
Line 1: $3x + y = 9$
When $x = 0 \implies y = 9 \implies (0, 9)$
When $y = 0 \implies x = 3 \implies (3, 0)$
Testing $(0, 0)$: $3(0) + 0 = 0 \leq 9$ (holds true, so the half-plane contains the origin).
Line 2: $x + 2y = 8$ When $x = 0 \implies y = 4 \implies (0, 4)$ When $y = 0 \implies x = 8 \implies (8, 0)$ Testing $(0, 0)$: $0 + 2(0) = 0 \leq 8$ (holds true, so the half-plane contains the origin).
Intersection point of boundary lines: Solving the simultaneous linear equations:

$$\begin{aligned} 3x + y = 9 &\implies y = 9 - 3x \\ x + 2(9 - 3x) &= 8 \\ x + 18 - 6x &= 8 \\ -5x &= -10 \implies x = 2 \end{aligned}$$

Substituting $x = 2$:

$$y = 9 - 3(2) = 3$$

The intersection point is $B(2, 3)$.
The feasible region bounded by the coordinate axes and the lines is the convex polygon $OABC$ with vertices:

$$O(0, 0), \quad A(3, 0), \quad B(2, 3), \quad C(0, 4)$$

Corner points and objective function evaluation: Evaluating $Z = 40x + 50y$ at each extreme point:
At $O(0, 0)$:

$$Z = 40(0) + 50(0) = 0$$

At $A(3, 0)$:

$$Z = 40(3) + 50(0) = 120$$

At $B(2, 3)$:

$$Z = 40(2) + 50(3) = 80 + 150 = 230$$

At $C(0,4)$:

$$Z = 40(0) + 50(4) = 200$$

Conclusion: The maximum value of Z is Rs. 230, which occurs at the corner point $(2,3)$. Hence, the company should produce 2 units of type A and 3 units of type B to obtain a maximum profit of Rs. 230.

1.1.37 There are three coins. One is a two-headed coin, another is a biased coin that comes up heads 75% of the time and the third is an unbiased coin. One of the three coins is chosen at random and tossed. If it shows heads, what is the probability that it is the two-headed coin?

Solution: Let the events be defined as:

$$\begin{aligned} E_1 &: \text{The chosen coin is the two-headed coin} \\ E_2 &: \text{The chosen coin is the biased coin} \\ E_3 &: \text{The chosen coin is the unbiased coin} \\ A &: \text{The tossed coin shows heads} \end{aligned}$$

Prior probabilities: Since one coin is chosen uniformly at random among the three coins::

$$P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$$

Conditional probabilities of getting a head: For the two-headed coin, both faces are heads:

$$P(A \mid E_1) = 1$$

For the biased coin, it comes up heads 75% of the time:

$$P(A \mid E_2) = \frac{75}{100} = \frac{3}{4}$$

For the unbiased coin, heads and tails are equally likely:

$$P(A \mid E_3) = \frac{1}{2}$$

By the law of total probability, the marginal probability of observing a head is:

$$\begin{aligned} P(A) &= P(E_1)P(A \mid E_1) + P(E_2)P(A \mid E_2) + P(E_3)P(A \mid E_3) \\ &= \frac{1}{3}(1) + \frac{1}{3}\left(\frac{3}{4}\right) + \frac{1}{3}\left(\frac{1}{2}\right) = \frac{1}{3}\left(\frac{9}{4}\right) = \frac{3}{4} \end{aligned}$$

Posterior probability via Bayes' theorem: The conditional probability that the coin is the two-headed coin given that it showed heads is:

$$\begin{aligned} P(E_1 \mid A) &= \frac{P(E_1)P(A \mid E_1)}{P(A)} \\ &= \frac{\frac{1}{3} \cdot 1}{\frac{3}{4}} = \frac{1}{3} \times \frac{4}{3} = \frac{4}{9} \end{aligned}$$

The probability that it is the two-headed coin is $\frac{4}{9}$.

1.1.38 Find the vector and Cartesian equations of the plane passing through the points having position vectors $\mathbf{i}+\mathbf{j}-2\mathbf{k}$, $2\mathbf{i}-\mathbf{j}+\mathbf{k}$, $\mathbf{i}+2\mathbf{j}+\mathbf{k}$. Write the equation of a plane passing through the point $(2,3,7)$ and parallel to the plane obtained above. Hence, find the distance between the two parallel planes.

Solution: Let the given points be A, B, and C, with position vectors[cite: 1]:

$$\mathbf{a}=\mathbf{i}+\mathbf{j}-2\mathbf{k} \qquad \mathbf{b}=2\mathbf{i}-\mathbf{j}+\mathbf{k} \qquad \mathbf{c}=\mathbf{i}+2\mathbf{j}+\mathbf{k}$$

The displacement vectors lying in the plane are:

$$\mathbf{AB}=\mathbf{b}-\mathbf{a}=(2-1)\mathbf{i}+(-1-1)\mathbf{j}+(1-(-2))\mathbf{k}=\mathbf{i}-2\mathbf{j}+3\mathbf{k}$$
$$\mathbf{AC}=\mathbf{c}-\mathbf{a}=(1-1)\mathbf{i}+(2-1)\mathbf{j}+(1-(-2))\mathbf{k}=0\mathbf{i}+\mathbf{j}+3\mathbf{k}$$

A normal vector $\mathbf{n}$ to the plane is given by the cross product $\mathbf{AB}\times\mathbf{AC}$:

$$\mathbf{n}=\mathbf{AB}\times\mathbf{AC}=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 1 & -2 & 3\\ 0 & 1 & 3\end{vmatrix}=\mathbf{i}(-6-3)-\mathbf{j}(3-0)+\mathbf{k}(1-0)$$
$$=-9\mathbf{i}-3\mathbf{j}+\mathbf{k}$$

Multiplying by -1 to choose an outward normal with positive leading coordinate:

$$\mathbf{n}_1=9\mathbf{i}+3\mathbf{j}-\mathbf{k}$$

The vector equation of a plane passing through $\mathbf{a}$ with normal $\mathbf{n}_1$ is:

$$\mathbf{r}\cdot\mathbf{n}_1=\mathbf{a}\cdot\mathbf{n}_1$$

Evaluating the scalar product on the right-hand side:

$$\mathbf{a}\cdot\mathbf{n}_1=(\mathbf{i}+\mathbf{j}-2\mathbf{k})\cdot(9\mathbf{i}+3\mathbf{j}-\mathbf{k})=9+3+2=14$$

Thus, the vector equation of the plane is:

$$\mathbf{r}\cdot(9\mathbf{i}+3\mathbf{j}-\mathbf{k})=14$$

Substituting $\mathbf{r}=x\mathbf{i}+y\mathbf{j}+z\mathbf{k}$, the Cartesian equation of the plane is:

$$9x+3y-z=14$$

Equation of the parallel plane: Any plane parallel to $9x+3y-z=14$ has the same normal vector $\mathbf{n}_1=9\mathbf{i}+3\mathbf{j}-\mathbf{k}$, so its equation is of the form:

$$\mathbf{r}\cdot(9\mathbf{i}+3\mathbf{j}-\mathbf{k})=d \quad \text{or} \quad 9x+3y-z=d$$

Since this plane passes through the point $(2,3,7)$ with position vector $\mathbf{a}_2=2\mathbf{i}+3\mathbf{j}+7\mathbf{k}$:

$$d=9(2)+3(3)-(7)=18+9-7=20$$

Parallel plane equations:

Vector equation:

$$\mathbf{r} \cdot (9\mathbf{i} + 3\mathbf{j} - \mathbf{k}) = 20$$

Cartesian equation:

$$9x + 3y - z = 20$$

The Cartesian equations of the two parallel planes are:

$$\Pi_1 : 9x + 3y - z - 14 = 0$$
$$\Pi_2 : 9x + 3y - z - 20 = 0$$

The perpendicular distance D between two parallel planes $Ax + By + Cz + D_1 = 0$ and $Ax + By + Cz + D_2 = 0$ is given by:

$$D = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}}$$

Substituting $A = 9$, $B = 3$, $C = -1$, $D_1 = -14$, and $D_2 = -20$:

$$D = \frac{|-14 - (-20)|}{\sqrt{9^2 + 3^2 + (-1)^2}} = \frac{|-14 + 20|}{\sqrt{81 + 9 + 1}} = \frac{6}{\sqrt{91}} \text{ units}$$

1.1.39 Find the equation of the line passing through $(2, -1, 2)$ and $(5, 3, 4)$ and of the plane passing through $(2, 0, 3)$, $(1, 1, 5)$ and $(3, 2, 4)$. Also, find their point of intersection. **Solution:** Equation of the line: Let the two given points on the line be $P(2, -1, 2)$ and $Q(5, 3, 4)$[cite: 1]. The direction ratios of the line passing through P and Q are:

$$a = 5 - 2 = 3 \qquad b = 3 - (-1) = 4 \qquad c = 4 - 2 = 2$$

The vector equation of a line passing through a point with position vector $\mathbf{a}_1 = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ along the direction vector $\mathbf{b}_1 = 3\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$ is:

$$\mathbf{r} = (2\mathbf{i} - \mathbf{j} + 2\mathbf{k}) + \lambda(3\mathbf{i} + 4\mathbf{j} + 2\mathbf{k})$$

The Cartesian equation of the line is:

$$\frac{x - 2}{3} = \frac{y + 1}{4} = \frac{z - 2}{2} = \lambda$$

Equation of the plane: Let the three given points on the plane be $A(2, 0, 3)$, $B(1, 1, 5)$, and $C(3, 2, 4)$[cite: 1]. The vectors lying along the plane are:

$$\mathbf{AB} = (1 - 2)\mathbf{i} + (1 - 0)\mathbf{j} + (5 - 3)\mathbf{k} = -\mathbf{i} + \mathbf{j} + 2\mathbf{k}$$
$$\mathbf{AC} = (3 - 2)\mathbf{i} + (2 - 0)\mathbf{j} + (4 - 3)\mathbf{k} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$$

A normal vector $\mathbf{n}$ to the plane is given by the vector cross product $\mathbf{AB} \times \mathbf{AC}$:

$$\begin{aligned}\mathbf{n} = \mathbf{AB} \times \mathbf{AC} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 1 & 2 \\ 1 & 2 & 1 \end{vmatrix} \\ &= \mathbf{i}\left[(1)(1) - (2)(2)\right] - \mathbf{j}\left[(-1)(1) - (2)(1)\right] + \mathbf{k}\left[(-1)(2) - (1)(1)\right] \\ &= \mathbf{i}(1-4) - \mathbf{j}(-1-2) + \mathbf{k}(-2-1) = -3\mathbf{i} + 3\mathbf{j} - 3\mathbf{k}\end{aligned}$$

Dividing by -3, an equivalent unit-scaled normal vector is:

$$\mathbf{n}_0 = \mathbf{i} - \mathbf{j} + \mathbf{k}$$

The vector equation of the plane passing through $A(2,0,3)$ with position vector $\mathbf{a} = 2\mathbf{i} + 0\mathbf{j} + 3\mathbf{k}$ is:

$$\mathbf{r} \cdot \mathbf{n}_0 = \mathbf{a} \cdot \mathbf{n}_0$$

Evaluating the scalar product on the right-hand side:

$$\mathbf{a} \cdot \mathbf{n}_0 = (2\mathbf{i} + 0\mathbf{j} + 3\mathbf{k}) \cdot (\mathbf{i} - \mathbf{j} + \mathbf{k}) = 2(1) + 0(-1) + 3(1) = 5$$

Thus, the vector equation of the plane is:

$$\mathbf{r} \cdot (\mathbf{i} - \mathbf{j} + \mathbf{k}) = 5$$

Substituting $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, the Cartesian equation of the plane is:

$$x - y + z = 5$$

Point of intersection: Any general point on the line can be parameterized in terms of scalar λ as:

$$x = 3\lambda + 2 \qquad y = 4\lambda - 1 \qquad z = 2\lambda + 2$$

Substituting these coordinates into the Cartesian equation of the plane:

$$(3\lambda + 2) - (4\lambda - 1) + (2\lambda + 2) = 5 \implies 3\lambda + 2 - 4\lambda + 1 + 2\lambda + 2 = 5 \implies \lambda = 0$$

Substituting $\lambda = 0$ back into the parametric coordinates:

$$x = 3(0) + 2 = 2 \qquad y = 4(0) - 1 = -1 \qquad z = 2(0) + 2 = 2$$

Therefore, the point of intersection of the line and the plane is $(2, -1, 2)$.

B. Class-10

1.2.1 Write the coordinates of the point P on x-axis which is equidistant from the points A(-2,0) and B(6,0).

Solution: Any point on the x-axis has its y-coordinate equal to 0. Let the coordinates of point P be $(x, 0)$.

Since point P is equidistant from $A(-2, 0)$ and $B(6, 0)$:

$$PA = PB \implies PA^2 = PB^2 \tag{1.2.1.1}$$

Applying the distance formula:

$$[x-(-2)]^2+(0-0)^2=(x-6)^2+(0-0)^2 \implies (x+2)^2=(x-6)^2 \tag{1.2.1.2}$$

Expanding both sides:

$$x^2+4x+4=x^2-12x+36 \implies 4x+4=-12x+36 \implies x=2 \tag{1.2.1.3}$$

Therefore, the coordinates of point P are $(2,0)$.

1.2.2 Evaluate :

$$\sin^2(60^\circ)+2\tan(45^\circ)-\cos^2(30^\circ)$$

Solution: Substituting the trignometric values into the given expression:

$$\begin{aligned}\sin^2(60^\circ)+2\tan(45^\circ)-\cos^2(30^\circ) &= \left(\frac{\sqrt{3}}{2}\right)^2+2(1)-\left(\frac{\sqrt{3}}{2}\right)^2\\ &= \frac{3}{4}+2-\frac{3}{4}=2\end{aligned}$$

Therefore, the evaluated value is **2**.

1.2.3 If $\sin A=\frac{3}{4}$,calculate $\sec A$

Solution: Using the trigonometric identity:

$$\cos^2 A=1-\sin^2 A$$

Substituting the given value of sin A:

$$\cos^2 A=\frac{7}{16} \qquad \cos A=\frac{\sqrt{7}}{4} \qquad \sec A=\frac{4}{\sqrt{7}}$$

1.2.4 In Figure 1, ABC is an isosceles triangle right angled at C with AC = 4 cm. Find the length of AB.

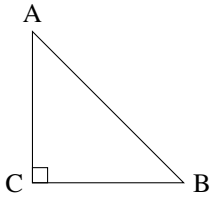


Figure 1

Solution: Triangle is a right-angled triangle at C and is isoceles.Because hypotenuse AB is the longest side, the two equal legs are AC and BC:

$$BC=AC=4 \text{ cm}$$

Applying the Pythagoras theorem in right-angled triangle ABC:

$$AB^2=AC^2+BC^2 \implies AB^2=4^2+4^2 \implies AB=\sqrt{32} \implies AB=4\sqrt{2} \text{ cm}$$

Therefore, the length of AB is:

$$AB = 4\sqrt{2} \text{ cm}$$

1.2.5 In Figure 2,DE $\parallel$ BC.Find the length of side AD,given that AE = 1·8 cm,BD = 7·2 cm and CE = 5·4 cm.

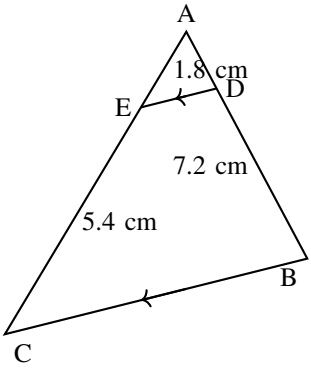


Figure 2

Solution: In triangle ABC, it is given that DE is parallel to BC. $DE \parallel BC$
By the Basic Proportionality Theorem (Thales' Theorem):

$$\frac{AD}{BD} = \frac{AE}{CE}$$

Substituting the given values[cite: 5]:

$$\frac{AD}{7.2} = \frac{1.8}{5.4} \implies \frac{AD}{7.2} = \frac{1}{3} \implies AD = \frac{7.2}{3} \implies AD = 2.4 \text{ cm}$$

Therefore, the length of side AD is:

$$AD = 2.4 \text{ cm}$$

1.2.6 Find the nature of roots of the quadratic equation $2x^2 - 4x + 3 = 0$.
Solution: We obtain[:

$$a = 2 \qquad b = -4 \qquad c = 3$$

The discriminant is defined as:

$$D = b^2 - 4ac$$

Substituting the values of a, b, and c:

$$D = (-4)^2 - 4(2)(3) = 16 - 24 = -8$$

Since the discriminant is less than zero:

$$D < 0$$

Therefore, the given quadratic equation has no real roots. NEED TO ADD GRAPH

AND PYTHON CODE

1.2.7 If HCF (336, 54) = 6, find LCM (336, 54).

Solution: Let the two given numbers be: a $= 336$ b $= 54$

$$\text{HCF}(a, b) = 6$$

Using the property relating the HCF and LCM of two numbers:

$$\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$$

Rearranging the equation to solve for the LCM:

$$\text{LCM}(a, b) = \frac{a \times b}{\text{HCF}(a, b)}$$

$$\text{LCM}(336, 54) = \frac{336 \times 54}{6} = 336 \times 9 = 3024$$

Therefore, the value of the LCM is:

$$\text{LCM}(336, 54) = 3024$$

1.2.8 Find the 21st term of the A.P.$-4\frac{1}{2},\ -3,\ -1\frac{1}{2},\ \ldots$

Solution: The common difference d is:

$$d = a_2 - a$$

$$= -3 - \left(-\frac{9}{2}\right) = -3 + \frac{9}{2} = \frac{-6+9}{2} = \frac{3}{2}$$

The general formula for the n-th term of an arithmetic progression is:

$$a_n = a + (n-1)d$$

Substituting n = 21 to find the 21st term:

$$a_{21} = -\frac{9}{2} + (21-1)\left(\frac{3}{2}\right)$$

$$= -\frac{9}{2} + 20\left(\frac{3}{2}\right) = -\frac{9}{2} + 30 = \frac{-9+60}{2} = \frac{51}{2}$$

Therefore, the 21st term of the A.P. is:

$$a_{21} = \frac{51}{2} \quad \left(\text{or } 25\frac{1}{2}\right)$$

1.2.9 For what value of k will have infinitely many solutions for the equations $2x+3y = 7$ and $(k+2)x - 3(1-k)y = 5k+1$.

Solution:

Form the augmented matrix for the linear network:

$$[\mathbf{A} \mid \mathbf{b}] = \left[\begin{array}{cc|c} 2 & 3 & 7 \\ k+2 & 3k-3 & 5k+1 \end{array}\right]$$

Perform the elementary row operation:

$$R_2 \leftarrow 2R_2 - (k+2)R_1$$

Evaluating the elements of the second row:

$$\begin{aligned} \text{First column: } & 2(k+2) - 2(k+2) = 0 \\ \text{Second column: } & 2(3k-3) - 3(k+2) = 6k - 6 - 3k - 6 = 3k - 12 \\ \text{Augmented column: } & 2(5k+1) - 7(k+2) = 10k + 2 - 7k - 14 = 3k - 12 \end{aligned}$$

The matrix reduces to:

$$\left[\begin{array}{cc|c} 2 & 3 & 7 \\ 0 & 3k-12 & 3k-12 \end{array}\right]$$

For the system to have infinitely many solutions, the second row must become a row of complete zeros (creating a free variable and ensuring consistency):

$$3k - 12 = 0 \implies 3k = 12 \implies k = 4$$

1.2.10 Find the mode of the following frequency distribution:

Class Interval	25–30	30–35	35–40	40–45	45–50	50–55
Frequency	25	34	50	42	38	14

Solution: The maximum frequency is 50, which lies in the class interval 35–40. Therefore, the modal class is 35–40[cite: 5].
From the modal class:

$$\begin{aligned} \text{Lower limit, } l &= 35 \\ \text{Class size, } h &= 40 - 35 = 5 \\ \text{Frequency of modal class, } f_1 &= 50 \\ \text{Frequency of class preceding modal class, } f_0 &= 34 \\ \text{Frequency of class succeeding modal class, } f_2 &= 42 \end{aligned}$$

The formula for the mode of grouped data is:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

Substituting the values:

$$\begin{aligned}\text{Mode} &= 35 + \left(\frac{50-34}{2(50)-34-42}\right) \times 5 \\ &= 35 + \left(\frac{16}{24}\right) \times 5 \\ &= 35 + \frac{10}{3} \\ &= 38.33\end{aligned}$$

1.2.11 The larger of two supplementary angles exceeds the smaller by 18°. Find the angles.
Solution: Let the larger angle be x and the smaller angle be y.
Since the two angles are supplementary, their sum is 180 degrees:

$$x + y = 180^\circ$$

The larger angle exceeds the smaller by 18 degrees

$$x - y = 18^\circ$$

Adding both equations:

$$\begin{aligned}(x+y) + (x-y) &= 180^\circ + 18^\circ \\ 2x &= 198^\circ \implies x = 99^\circ\end{aligned}$$

Substituting the value of x into the first equation:

$$99^\circ + y = 180^\circ \implies y = 81^\circ$$

Therefore, the two supplementary angles are:

$$x = 99^\circ \qquad\qquad y = 81^\circ$$

1.2.12 Sumit is 3 times as old as his son. Five years later, he shall be two and a half times as old as his son. How old is Sumit at present?
Solution: Let the present age of the son be: x years
Then, the present age of Sumit is: y = 3x
Five years later:

$$\begin{aligned}\text{Son's age} &= x + 5 \\ \text{Sumit's age} &= y + 5\end{aligned}$$

According to the given condition, Sumit will be two and a half times as old as his son:

$$y + 5 = \frac{5}{2}(x+5) \implies 2y + 10 \qquad\qquad = 5x + 25$$

Substituting y = 3x into the equation:

$$\begin{aligned} 2(3x) + 10 &= 5x + 25 \\ 6x - 5x = 25 - 10 &\implies x = 15 \end{aligned}$$

Now, finding Sumit's present age:

$$\begin{aligned} y &= 3x \\ y &= 45 \end{aligned}$$

1.2.13 The probability of selecting a blue marble at random from a jar that contains only blue, black and green marbles is $\frac{1}{5}$. The probability of selecting a black marble at random from the same jar is $\frac{1}{4}$. If the jar contains 11 green marbles, find the total number of marbles in the jar.

Solution: Let the total number of marbles in the jar be n.

Since the jar contains only blue, black, and green marbles, the sum of probabilities of all possible outcomes is equal to 1:

$$P(\text{blue}) + P(\text{black}) + P(\text{green}) = 1$$

Substituting these values to determine the probability of selecting a green marble:

$$\begin{aligned} \frac{1}{5} + \frac{1}{4} + P(\text{green}) &= 1 \\ \frac{9}{20} + P(\text{green}) = 1 &\implies P(\text{green}) = \frac{11}{20} \end{aligned}$$

The probability of selecting a green marble is also given by the ratio of green marbles to total marbles:

$$P(\text{green}) = \frac{\text{Number of green marbles}}{\text{Total number of marbles}} \implies \frac{11}{20} = \frac{11}{n} \implies n = 20$$

1.2.14 Find a relation between x and y if the points $A(x, y)$, $B(-4, 6)$ and $C(-2, 3)$ are collinear.

Solution: Construct the displacement vectors with point A as the reference origin:

$$\begin{aligned} \mathbf{v}_1 = \mathbf{AB} &= \begin{bmatrix} -4 - x \\ 6 - y \end{bmatrix} \\ \mathbf{v}_2 = \mathbf{BC} &= \begin{bmatrix} -2 - (-4) \\ 3 - 6 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix} \end{aligned}$$

For collinearity, the vectors must be linearly dependent, meaning the column matrix formed by them has a rank of 1 and its determinant is zero:

$$\begin{aligned} \det \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix} &= 0 \\ \det \begin{bmatrix} -4 - x & 2 \\ 6 - y & -3 \end{bmatrix} &= 0 \\ (-4 - x)(-3) - (2)(6 - y) = 0 &\implies 3x + 2y = 0 \end{aligned}$$

1.2.15 Find the area of a triangle whose vertices are $(1, -1)$, $(-4, 6)$ and $(-3, -5)$.

Solution: Taking the first vertex as the origin, form two displacement edge vectors:

$$\mathbf{u} = \mathbf{AB} = \begin{bmatrix} -4-1 \\ 6-(-1) \end{bmatrix} = \begin{bmatrix} -5 \\ 7 \end{bmatrix}$$
$$\mathbf{v} = \mathbf{AC} = \begin{bmatrix} -3-1 \\ -5-(-1) \end{bmatrix} = \begin{bmatrix} -4 \\ -4 \end{bmatrix}$$

The area corresponds to half the absolute value of the 2x2 determinant formed by these column vectors:

$$\begin{aligned} \text{Area} &= \frac{1}{2} \left| \det \begin{bmatrix} \mathbf{u} & \mathbf{v} \end{bmatrix} \right| \\ &= \frac{1}{2} \left| \det \begin{bmatrix} -5 & -4 \\ 7 & -4 \end{bmatrix} \right| \\ &= \frac{1}{2} |48| = 24 \text{ sq. units} \end{aligned}$$

1.2.16 Write the smallest number which is divisible by both 306 and 657.

Solution: Compute the HCF of the two numbers using Euclid's division lemma:

$$\begin{aligned} 657 &= 306 \times 2 + 45 \\ 306 &= 45 \times 6 + 36 \\ 45 &= 36 \times 1 + 9 \\ 36 &= 9 \times 4 + 0 \end{aligned}$$

Since the remainder is 0, the HCF is:

$$\text{HCF}(306, 657) = 9$$

Using the property relating the HCF and LCM of two numbers:

$$\begin{aligned} \text{LCM}(a, b) &= \frac{a \times b}{\text{HCF}(a, b)} \\ \text{LCM}(306, 657) &= \frac{306 \times 657}{9} \\ &= 34 \times 657 \\ &= 22338 \end{aligned}$$

1.2.17 Point A lies on the line segment XY joining $X(6, -6)$ and $Y(-4, -1)$ in such a way that $\frac{XA}{XY} = \frac{2}{5}$. If point A also lies on the line $3x + k(y + 1) = 0$.find the value of k.

Solution: Point A lies on the line segment XY such that[cite: 6]:

$$\begin{aligned}\frac{XA}{XY} &= \frac{2}{5}\\ \frac{XA}{XA+AY} &= \frac{2}{5}\\ 5XA &= 2(XA+AY)\\ 3XA &= 2AY\\ \frac{XA}{AY} &= \frac{2}{3}\end{aligned}$$

Thus, point A divides the segment joining X(6, -6) and Y(-4, -1) internally in the ratio:

$$m : n = 2 : 3$$

Using the section formula for coordinates of point A:

$$\begin{aligned}x &= \frac{mx_2 + nx_1}{m+n}\\ &= \frac{2(-4)+3(6)}{2+3}\\ &= \frac{10}{5} = 2\end{aligned}$$

$$\begin{aligned}y &= \frac{my_2 + ny_1}{m+n}\\ &= \frac{2(-1)+3(-6)}{2+3}\\ &= \frac{-20}{5} = -4\end{aligned}$$

Therefore, the coordinates of point A are:

$$A(2, -4)$$

Since point A(2, -4) lies on the given line:

$$\begin{aligned}3x + k(y+1) &= 0\\ 3(2) + k(-4+1) &= 0\\ 6 - 3k = 0 &\implies k = 2\end{aligned}$$

1.2.18 Solve for x: $x^2 + 5x - (a^2 + a - 6) = 0$.

Solution: The equation with the standard quadratic form $Ax^2 + Bx + C = 0$

$$A = 1 \qquad B = 5 \qquad C = -(a^2 + a - 6)$$

The discriminant D is:

$$\begin{aligned} D &= B^2 - 4AC \\ &= 5^2 - 4(1)\left[-(a^2 + a - 6)\right] \\ &= 4a^2 + 4a + 1 = (2a+1)^2 \end{aligned}$$

Using the quadratic formula:

$$\begin{aligned} x &= \frac{-B \pm \sqrt{D}}{2A} \\ &= \frac{-5 \pm \sqrt{(2a+1)^2}}{2(1)} \\ &= \frac{-5 \pm (2a+1)}{2} \end{aligned}$$

Taking the positive sign:

$$\begin{aligned} x &= \frac{-5 + (2a+1)}{2} \\ &= \frac{2a-4}{2} = a - 2 \end{aligned}$$

Taking the negative sign:

$$\begin{aligned} x &= \frac{-5 - (2a+1)}{2} \\ &= \frac{-2a-6}{2} = -(a+3) \end{aligned}$$

Therefore, the values of x are:

$$x = a - 2 \quad \text{(or)} \quad x = -(a+3)$$

1.2.19 Find A and B if $\sin(A + 2B) = \frac{\sqrt{3}}{2}$ and $\cos(A + 4B) = 0$, where A and B are acute angles.

Solution: where A and B are acute angles[cite: 1]. *(Note: Corrected from $\frac{3}{2}$ to $\frac{\sqrt{3}}{2}$ since the range of the sine function for real angles is $[-1, 1]$)*[cite: 1].

Solution:

Since $\sin(60^\circ) = \frac{\sqrt{3}}{2}$ and the angles are acute:

$$A + 2B = 60^\circ$$

Since $\cos(90^\circ) = 0$:

$$A + 4B = 90^\circ$$

Writing the system of linear equations in matrix form $\mathbf{Mx} = \mathbf{b}$:

$$\begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 60^\circ \\ 90^\circ \end{bmatrix}$$

The determinant of the system matrix $\mathbf{M}$ is:

$$\det(\mathbf{M}) = (1)(4) - (2)(1) = 2$$

The inverse matrix $\mathbf{M}^{-1}$ is:

$$\mathbf{M}^{-1} = \frac{1}{2}\begin{bmatrix} 4 & -2 \\ -1 & 1 \end{bmatrix}$$

Solving for state vector $\mathbf{x}$:

$$\begin{aligned}\begin{bmatrix} A \\ B \end{bmatrix} &= \frac{1}{2}\begin{bmatrix} 4 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 60^\circ \\ 90^\circ \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} 4(60^\circ) - 2(90^\circ) \\ -1(60^\circ) + 1(90^\circ) \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 60^\circ \\ 30^\circ \end{bmatrix} = \begin{bmatrix} 30^\circ \\ 15^\circ \end{bmatrix}\end{aligned}$$

Therefore, the values of A and B are:

$$A = 30^\circ \qquad\qquad B = 15^\circ$$

1.2.20 Prove that $2 + 5\sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is an irrational number.
Solution: Assume to the contrary that $2 + 5\sqrt{3}$ is a rational number.
Then, it can be expressed in the form $\frac{a}{b}$, where a and b are co-prime integers and $b \neq 0$:

$$2 + 5\sqrt{3} = \frac{a}{b}$$

Rearranging the terms:

$$\begin{aligned}5\sqrt{3} &= \frac{a}{b} - 2 \\ 5\sqrt{3} &= \frac{a - 2b}{b} \\ \sqrt{3} &= \frac{a - 2b}{5b}\end{aligned}$$

Since a and b are integers, $a - 2b$ and $5b$ are also integers ($5b \neq 0$). Thus, the right-hand side represents a rational number:

$$\frac{a - 2b}{5b} \in \mathbb{Q}$$

This implies:

$$\sqrt{3} \in \mathbb{Q}$$

This contradicts the given fact that $\sqrt{3}$ is an irrational number[cite: 1].
Therefore, our assumption that $2+5\sqrt{3}$ is rational is false, and $2+5\sqrt{3}$ is an irrational number.

1.2.21 Using Euclid's Algorithm, find the HCF of 2048 and 960.
Solution: The Euclidean algorithm corresponds to applying elementary unimodular shear matrices to reduce an integer state vector to its principal generator in a 1D

lattice:

$$\mathbf{v}_0 = \begin{bmatrix} 2048 \\ 960 \end{bmatrix}$$

Applying the first shear transformation $\mathbf{E}_1 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$:

$$\begin{aligned} \mathbf{v}_1 &= \mathbf{E}_1 \mathbf{v}_0 \\ &= \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2048 \\ 960 \end{bmatrix} = \begin{bmatrix} 2048 - 2(960) \\ 960 \end{bmatrix} = \begin{bmatrix} 128 \\ 960 \end{bmatrix} \end{aligned}$$

Applying the second shear transformation $\mathbf{E}_2 = \begin{bmatrix} 1 & 0 \\ -7 & 1 \end{bmatrix}$:

$$\begin{aligned} \mathbf{v}_2 &= \mathbf{E}_2 \mathbf{v}_1 \\ &= \begin{bmatrix} 1 & 0 \\ -7 & 1 \end{bmatrix} \begin{bmatrix} 128 \\ 960 \end{bmatrix} = \begin{bmatrix} 128 \\ 960 - 7(128) \end{bmatrix} = \begin{bmatrix} 128 \\ 64 \end{bmatrix} \end{aligned}$$

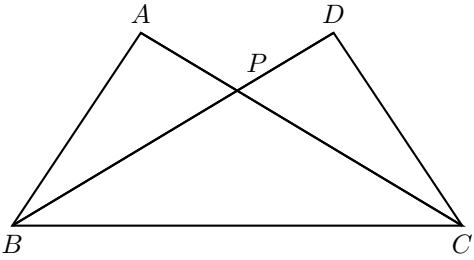
Applying the third shear transformation $\mathbf{E}_3 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$:

$$\begin{aligned} \mathbf{v}_3 &= \mathbf{E}_3 \mathbf{v}_2 \\ &= \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 128 \\ 64 \end{bmatrix} = \begin{bmatrix} 128 - 2(64) \\ 64 \end{bmatrix} = \begin{bmatrix} 0 \\ 64 \end{bmatrix} \end{aligned}$$

The non-zero invariant entry of the completely reduced state vector yields the greatest common divisor:

$$\gcd(2048, 960) = 64$$

1.2.22 Two right triangles ABC and DBC are drawn on the same hypotenuse BC and on the same side of BC. If AC and BD intersect at P, prove that $AP \times PC = BP \times DP$.



Solution: In right triangles ABC and DBC, both are right-angled on the same hypotenuse BC:

$$\angle BAC = 90^\circ \qquad\qquad \angle BDC = 90^\circ$$

Since hypotenuse BC subtends equal angles of 90° at vertices A and D on the same side, the four points A, B, C, and D are concyclic, lying on a circle with diameter

BC[cite: 1].
For the chord AD, the angles subtended in the same segment of the circle are equal:

$$\angle ABD = \angle ACD \qquad\qquad \angle PBA = \angle PCD$$

Now, compare $\triangle PAB$ and $\triangle PDC$:

$$\angle APB = \angle DPC \quad \text{(vertically opposite angles)}$$
$$\angle PBA = \angle PCD \quad \text{(angles in the same segment)}$$

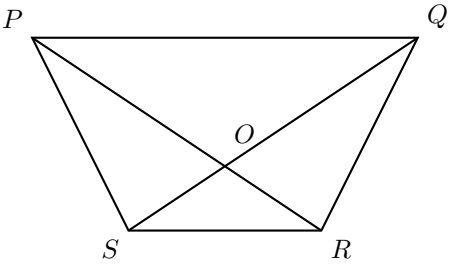
By the AA (Angle-Angle) similarity criterion:

$$\triangle PAB \sim \triangle PDC$$

Since the ratio of corresponding sides of similar triangles is equal:

$$\frac{AP}{DP} = \frac{BP}{PC} \implies AP \times PC = BP \times DP$$

1.2.23 Diagonals of a trapezium $PQRS$ intersect each other at the point O, $PQ \parallel RS$ and $PQ = 3RS$. Find the ratio of the areas of triangles POQ and ROS.



Solution: In trapezium $PQRS$, it is given that $PQ \parallel RS$[cite: 1].
In $\triangle POQ$ and $\triangle ROS$:

$$\angle POQ = \angle ROS \quad \text{(vertically opposite angles)}$$
$$\angle OPQ = \angle ORS \quad \text{(alternate interior angles, as } PQ \parallel RS\text{)}$$
$$\angle OQP = \angle OSR \quad \text{(alternate interior angles, as } PQ \parallel RS\text{)}$$

Therefore, by the AAA similarity criterion:

$$\triangle POQ \sim \triangle ROS$$

The ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides:

$$\frac{\text{Area}(\triangle POQ)}{\text{Area}(\triangle ROS)} = \left(\frac{PQ}{RS}\right)^2$$

Given that $PQ = 3RS$:

$$\frac{\text{Area}(\triangle POQ)}{\text{Area}(\triangle ROS)} = \left(\frac{3RS}{RS}\right)^2 = (3)^2 = \frac{9}{1}$$

1.2.24 In Figure 3, PQ and RS are two parallel tangents to a circle with centre O and another tangent AB with point of contact C intersects PQ at A and RS at B. Prove that $\angle AOB = 90°$.

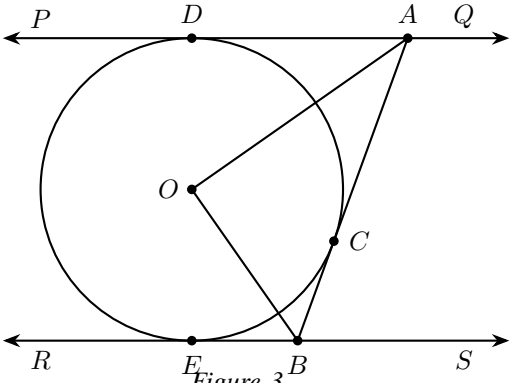


Figure 3

Solution: Let O be the origin $(0, 0)$. With tangent hyperplanes $y = r$ (PQ) and $y = -r$ (RS), parameterize contact point C on circle $\|\mathbf{x}\| = r$ by $\theta \in (-\pi/2, \pi/2)$:

$$\mathbf{r}_C = \begin{bmatrix} r\cos\theta \\ r\sin\theta \end{bmatrix}$$

The tangent hyperplane normal at C is $\hat{\mathbf{n}} = [\cos\theta, \sin\theta]^T$, yielding line equation $x\cos\theta + y\sin\theta = r$. Intersecting with boundaries $y = \pm r$:

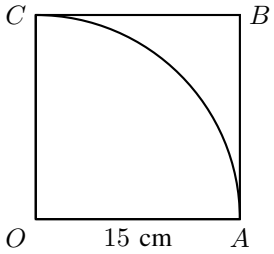
$$\mathbf{r}_A = \begin{bmatrix} r\frac{1-\sin\theta}{\cos\theta} \\ r \end{bmatrix}, \quad \mathbf{r}_B = \begin{bmatrix} r\frac{1+\sin\theta}{\cos\theta} \\ -r \end{bmatrix}$$

Evaluating the inner product:

$$\langle \mathbf{r}_A, \mathbf{r}_B \rangle = r^2 \left(\frac{1 - \sin^2\theta}{\cos^2\theta} \right) - r^2 = r^2(1) - r^2 = 0$$

Since $\langle \mathbf{r}_A, \mathbf{r}_B \rangle = 0$, position vectors $\mathbf{r}_A \perp \mathbf{r}_B \implies \angle AOB = 90°$.

1.2.25 In Figure 4, a square $OABC$ is inscribed in a quadrant $OPBQ$. If $OA = 15$ cm, find the area of the shaded region. (Use $\pi = 3.14$.)



Solution: Let the side of the square $OABC$ be:

$$s = OA = AB = 15 \text{ cm}$$

In $\triangle OAB$, $\angle OAB = 90°$. Diagonal OB forms the radius r of the circular quadrant $OPBQ$. Applying the Pythagoras theorem in right-angled triangle OAB:

$$OB^2 = OA^2 + AB^2$$
$$OB^2 = 15^2 + 15^2 \implies OB = \sqrt{450} = 15\sqrt{2} \text{ cm}$$

Thus, the radius of the quadrant is:

$$r = 15\sqrt{2} \text{ cm}$$

The area of quadrant $OPBQ$ is:

$$\text{Area of quadrant} = \frac{1}{4}\pi r^2 = \frac{1}{4} \times 3.14 \times (15\sqrt{2})^2 = \frac{1413}{4} = 353.25 \text{ cm}^2$$

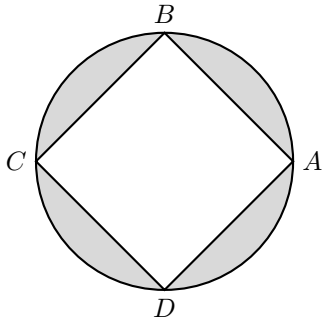
The area of square $OABC$ is:

$$\text{Area of square} = s^2 = 15^2 = 225 \text{ cm}^2$$

The area of the shaded region is given by:

$$\begin{aligned}\text{Area of shaded region} &= \text{Area of quadrant} - \text{Area of square} \\ &= 353.25 - 225 = 128.25 \text{ cm}^2\end{aligned}$$

1.2.26 In Figure 5, $ABCD$ is a square with side $2\sqrt{2}$ cm and inscribed in a circle. Find the area of the shaded region. (Use $\pi = 3.14$.)



Solution: Let the side of the inscribed square $ABCD$ be[cite: 2]:

$$a = 2\sqrt{2} \text{ cm}$$

Since each interior angle of a square is $90°$, diagonal AC subtends $90°$ at the circle's circumference, meaning AC is the diameter of the circumscribing circle:

$$AC^2 = AB^2 + BC^2$$
$$AC^2 = (2\sqrt{2})^2 + (2\sqrt{2})^2 \implies AC = \sqrt{16} = 4 \text{ cm}$$

The radius r of the circle is:

$$r = \frac{AC}{2} = \frac{4}{2} = 2 \text{ cm}$$

The area of the circumscribed circle is:

$$\text{Area of circle} = \pi r^2 = 3.14 \times 2^2 = 3.14 \times 4 = 12.56 \text{ cm}^2$$

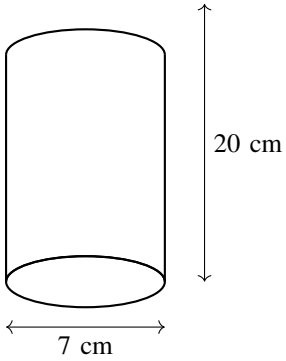
The area of the inscribed square is:

$$\text{Area of square} = a^2 = (2\sqrt{2})^2 = 8 \text{ cm}^2$$

The area of the shaded region is:

$$\begin{aligned}\text{Area of shaded region} &= \text{Area of circle} - \text{Area of square} \\ &= 12.56 - 8 = 4.56 \text{ cm}^2\end{aligned}$$

1.2.27 A solid is in the form of a cylinder with hemispherical ends. The total height of the solid is 20 cm and the diameter of the cylinder is 7 cm. Find the total volume of the solid. (Use $\pi = \frac{22}{7}$.)



Solution: The solid comprises a central right circular cylinder capped by two identical hemispherical ends.
From the given dimensions:

$$\text{Diameter, } d = 7 \text{ cm} \implies \text{Radius, } r = \frac{7}{2} \text{ cm}$$

$$\text{Total height, } H = 20 \text{ cm}$$

The height of the cylindrical portion h is obtained by subtracting the radii of both hemispherical ends:

$$h = H - 2r = 20 - 2\left(\frac{7}{2}\right) = 20 - 7 = 13 \text{ cm}$$

The volume of the cylindrical portion is:

$$\begin{aligned}V_{\text{cylinder}} &= \pi r^2 h \\ &= \frac{22}{7} \times \left(\frac{7}{2}\right)^2 \times 13 = \frac{22}{7} \times \frac{49}{4} \times 13 = \frac{11 \times 7 \times 13}{2} = \frac{1001}{2} = 500.5 \text{ cm}^3\end{aligned}$$

The two hemispherical ends combine to form a complete sphere of radius r:

$$V_{\text{ends}} = 2 \times \left(\frac{2}{3}\pi r^3\right) = \frac{4}{3}\pi r^3$$
$$= \frac{4}{3} \times \frac{22}{7} \times \left(\frac{7}{2}\right)^3 = \frac{4}{3} \times \frac{22}{7} \times \frac{343}{8} = \frac{11 \times 49}{3} = \frac{539}{3} \approx 179.67 \text{ cm}^3$$

Summing the component volumes:

$$V_{\text{total}} = V_{\text{cylinder}} + V_{\text{ends}}$$
$$= \frac{1001}{2} + \frac{539}{3} = \frac{3003 + 1078}{6} = \frac{4081}{6} = 680\frac{1}{6} \text{ cm}^3 \approx 680.17 \text{ cm}^3$$

1.2.28 The marks obtained by 100 students in an examination are given below:

Marks	30–35	35–40	40–45	45–50	50–55	55–60	60–65
Number of Students	14	16	28	23	18	8	3

Find the mean marks of the students.

Solution:

Marks	**Class Mark** (x_i)	**Frequency** (f_i)
30–35	32.5	14
35–40	37.5	16
40–45	42.5	28
50–55	52.5	18
55–60	57.5	8
60–65	62.5	3
Total		$\sum f_i = 110$

$$\sum f_i x_i = 14(32.5) + 16(37.5) + 28(42.5) + 23(47.5) + 18(52.5) + 8(57.5) + 3(62.5)$$
$$= 455 + 600 + 1190 + 1092.5 + 945 + 460 + 187.5 = 4930$$

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{4930}{110} = \frac{493}{11} \approx 44.82$$

1.2.29 For what value of k is the polynomial $f(x) = 3x^4 - 9x^3 + x^2 + 15x + k$ completely divisible by $3x^2 - 5$?

Solution: Work in the quotient ring $\mathbb{R}[x]/\langle 3x^2 - 5\rangle$, where the divisor induces the reduction relation:

$$3x^2 - 5 \equiv 0 \implies x^2 \equiv \frac{5}{3}$$

Higher powers of the state variable x reduce recursively:

$$x^3 = x \cdot x^2 \equiv \frac{5}{3}x \qquad\qquad x^4 = (x^2)^2 \equiv \left(\frac{5}{3}\right)^2 = \frac{25}{9}$$

Projecting $f(x)$ onto the 1st-degree polynomial subspace modulo $(3x^2-5)$:

$$f(x) \equiv 3\left(\frac{25}{9}\right) - 9\left(\frac{5}{3}x\right) + \left(\frac{5}{3}\right) + 15x + k \pmod{3x^2-5}$$
$$\equiv \frac{25}{3} - 15x + \frac{5}{3} + 15x + k \equiv 10 + k$$

Setting the residue projection to zero:

$$10 + k = 0 \implies k = -10$$

1.2.30 Find the zeroes of the quadratic polynomial $7y^2 - \frac{11}{3}y - \frac{2}{3}$ and verify the relationship between the zeroes and the coefficients.

Solution: Comparing $p(y) = 7y^2 - \frac{11}{3}y - \frac{2}{3}$ with standard quadratic form ay^2+by+c:

$$a = 7, \quad b = -\frac{11}{3}, \quad c = -\frac{2}{3}$$

Sum of zeroes:

$$\alpha + \beta = \frac{2}{3} + \left(-\frac{1}{7}\right) = \frac{14-3}{21} = \frac{11}{21} = -\frac{b}{a}$$

2. Product of zeroes:

$$\alpha\beta = \left(\frac{2}{3}\right)\left(-\frac{1}{7}\right) = -\frac{2}{21} = \frac{c}{a}$$

1.2.31 Prove that the ratio of the areas of two similar triangles is equal to the ratio of the squares on their corresponding sides.

Solution:

To Prove:

$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{CA}{RP}\right)^2$$

Proof:

The area of a triangle is given by half the product of its base and corresponding altitude:

$$\text{ar}(\triangle ABC) = \frac{1}{2} \times BC \times AM$$
$$\text{ar}(\triangle PQR) = \frac{1}{2} \times QR \times PN$$

Taking the ratio of their areas:

$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \frac{\frac{1}{2} \times BC \times AM}{\frac{1}{2} \times QR \times PN} = \left(\frac{BC}{QR}\right) \times \left(\frac{AM}{PN}\right)$$

In $\triangle ABM$ and $\triangle PQN$:

$$\angle B = \angle Q \quad \text{(corresponding angles of similar triangles, } \triangle ABC \sim \triangle PQR)$$
$$\angle AMB = \angle PNQ = 90^\circ \quad \text{(by construction)}$$

By the AA (Angle-Angle) similarity criterion:

$$\triangle ABM \sim \triangle PQN$$

Since corresponding sides of similar triangles are in proportion:

$$\frac{AM}{PN} = \frac{AB}{PQ}$$

Since $\triangle ABC \sim \triangle PQR$, their corresponding sides are proportional:

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP}$$

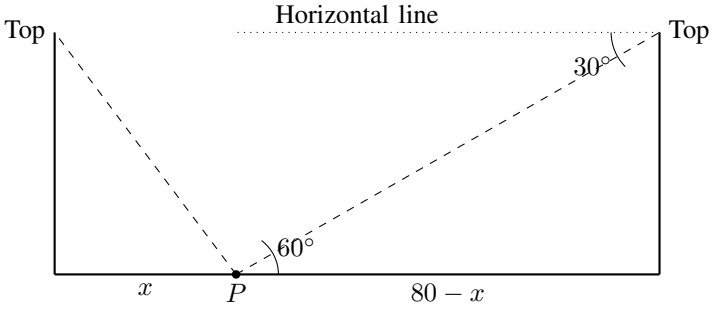
Substituting $\frac{AM}{PN} = \frac{BC}{QR}$ into the area ratio expression:

$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \left(\frac{BC}{QR}\right) \times \left(\frac{BC}{QR}\right) = \left(\frac{BC}{QR}\right)^2$$

Using the ratio equality of the corresponding sides:

$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{CA}{RP}\right)^2$$

1.2.32 Two poles of equal heights are standing opposite to each other on either side of the road which is 80 m wide. From a point P between them on the road, the angle of elevation of the top of a pole is 60° and the angle of depression from the top of the other pole to point P is 30°. Find the heights of the poles and the distance of the point P from the poles.



Road width = 80 m

Solution: From the two geometric projection constraints:

$$\sqrt{3}x - h = 0 \qquad\qquad x + \sqrt{3}h = 80$$

Formulating as a linear vector system $\mathbf{Av} = \mathbf{b}$:

$$\begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix} \begin{bmatrix} x \\ h \end{bmatrix} = \begin{bmatrix} 0 \\ 80 \end{bmatrix}$$

$$\mathbf{A} = 2\begin{bmatrix}\cos(30^\circ) & -\sin(30^\circ)\\ \sin(30^\circ) & \cos(30^\circ)\end{bmatrix}$$

The determinant is:

$$\det(\mathbf{A}) = (\sqrt{3})(\sqrt{3}) - (-1)(1) = 3 + 1 = 4$$

The inverse matrix is:

$$\mathbf{A}^{-1} = \frac{1}{4}\begin{bmatrix}\sqrt{3} & 1\\ -1 & \sqrt{3}\end{bmatrix}$$

Solving for the state coordinates:

$$\begin{bmatrix}x\\ h\end{bmatrix} = \frac{1}{4}\begin{bmatrix}\sqrt{3} & 1\\ -1 & \sqrt{3}\end{bmatrix}\begin{bmatrix}0\\ 80\end{bmatrix} = \frac{1}{4}\begin{bmatrix}80\\ 80\sqrt{3}\end{bmatrix} = \begin{bmatrix}20\\ 20\sqrt{3}\end{bmatrix}$$

Thus, $x = 20$ m and $h = 20\sqrt{3}$ m.

1.2.33 The total cost of a certain length of a piece of cloth is Rs. 200. If the piece was 5 m longer and each metre of cloth costs Rs. 2 less, the cost of the piece would have remained unchanged. How long is the piece and what is its original rate per metre?

Solution:

$$\text{Original rate} = \frac{200}{x}\ \text{Rs./m}$$

$$\text{New length} = (x+5)\text{m} \qquad \text{New rate} = \left(\frac{200}{x} - 2\right)\text{Rs./m}$$

The total cost remains unchanged at Rs. 200:

$$(\text{New length}) \times (\text{New rate}) = 200$$
$$(x+5)\left(\frac{200}{x} - 2\right) = 200$$

Expanding the expression:

$$x\left(\frac{200}{x}\right) - 2x + 5\left(\frac{200}{x}\right) - 10 = 200 \implies -2x + \frac{1000}{x} - 10 = 0$$
$$\implies -2x^2 - 10x + 1000 = 0 \implies x^2 + 5x - 500 = 0$$

Factorising the quadratic equation:

$$x^2 + 25x - 20x - 500 = 0 \implies (x-20)(x+25) = 0$$

Equating each factor to zero:

$$x - 20 = 0 \implies x = 20 \qquad x + 25 = 0 \implies x = -25$$

Since length cannot be negative, $x = 20$ m

Evaluating the original rate per metre:

$$\text{Original rate} = \frac{200}{20} = 10 \text{ Rs./m}$$

1.2.34 Which term of the Arithmetic Progression -7, -12, -17, -22, ...will be -82? Is -100 any term of the A.P.? Give reason for your answer.

Solution: For the given Arithmetic Progression $-7, -12, -17, -22, \ldots$:

$$\text{First term, } a = -7 \qquad \text{Common difference, } d = -12 - (-7) = -5$$

Let the nth term be -82:

$$a_n = a + (n-1)d$$
$$-82 = -7 + (n-1)(-5) \implies -75 = -5(n-1)$$
$$\implies n - 1 = \frac{-75}{-5} = 15 \implies n = 16$$

Now, suppose -100 is the mth term of the progression:

$$a_m = a + (m-1)d$$
$$-100 = -7 + (m-1)(-5)$$
$$-100 + 7 = -5(m-1)$$
$$m - 1 = \frac{93}{5} = 18.6 \implies m = 19.6$$

Since the term index m must be a positive integer ($m \in \mathbb{Z}^+$), -100 cannot be a term of this A.P.

1.2.35 How many terms of the Arithmetic Progression 45, 39, 33, ... must be taken so that their sum is 180? Explain the double answer.

Solution: For the given Arithmetic Progression $45, 39, 33, \ldots$[cite: 3]:

$$\text{First term, } a = 45$$
$$\text{Common difference, } d = 39 - 45 = -6$$

Let the sum of the first n terms be $S_n = 180$:

$$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$$
$$180 = \frac{n}{2}\left[2(45) + (n-1)(-6)\right] \implies 180 = n(48 - 3n) \implies 180 = 48n - 3n^2$$

Rearranging into standard quadratic form:

$$3n^2 - 48n + 180 = 0 \implies n^2 - 16n + 60 = 0$$

Factorising the quadratic equation:

$$n^2 - 10n - 6n + 60 = 0 \implies (n-6)(n-10) = 0$$

Equating each factor to zero:

$$n = 6 \quad \text{(or)} \quad n = 10$$

Explanation of the double answer:
The common difference is negative ($d = -6$), meaning the terms decrease and eventually become negative.
Evaluating the terms from $n = 7$ to $n = 10$:

$$\begin{aligned} a_7 &= 45 + 6(-6) = 9 \\ a_8 &= 45 + 7(-6) = 3 \\ a_9 &= 45 + 8(-6) = -3 \\ a_{10} &= 45 + 9(-6) = -9 \end{aligned}$$

The sum of these additional four terms is:

$$a_7 + a_8 + a_9 + a_{10} = 9 + 3 + (-3) + (-9) = 0$$

Since the sum of the 7th, 8th, 9th, and 10th terms is zero, adding them to the first 6 terms does not change the cumulative sum. Thus, both $n = 6$ and $n = 10$ give $S_n = 180$.

1.2.36 Prove that

$$\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\csc\theta.$$

Solution:

$$\text{LHS} = \frac{\tan\theta}{1-\frac{1}{\tan\theta}} + \frac{\frac{1}{\tan\theta}}{1-\tan\theta} = \frac{\tan^2\theta}{\tan\theta - 1} - \frac{1}{\tan\theta(\tan\theta - 1)}$$

$$\text{LHS} = \frac{\tan^3\theta - 1}{\tan\theta(\tan\theta - 1)}$$

Applying the algebraic cubic factorization $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$:

$$\begin{aligned} \text{LHS} &= \frac{(\tan\theta - 1)(\tan^2\theta + \tan\theta + 1)}{\tan\theta(\tan\theta - 1)} \\ &= \frac{\tan^2\theta + \tan\theta + 1}{\tan\theta} = \tan\theta + 1 + \cot\theta \end{aligned}$$

$$\text{LHS} = 1 + \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = 1 + \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta}$$

$$\text{LHS} = 1 + \frac{1}{\sin\theta\cos\theta} = 1 + \sec\theta\csc\theta = \text{RHS}$$

1.2.37 Prove that

$$\frac{\sin\theta}{\cot\theta+\csc\theta}=2+\frac{\sin\theta}{\cot\theta-\csc\theta}.$$

Solution: Rearranging the identity to bring the variable fractional terms together:

$$\frac{\sin\theta}{\cot\theta+\csc\theta}-\frac{\sin\theta}{\cot\theta-\csc\theta}=2$$

Evaluating the Left-Hand Side:

$$\text{LHS}=\frac{\sin\theta}{\csc\theta+\cot\theta}-\frac{\sin\theta}{-(\csc\theta-\cot\theta)}=\sin\theta\left[\frac{1}{\csc\theta+\cot\theta}+\frac{1}{\csc\theta-\cot\theta}\right]$$

$$\text{LHS}=\sin\theta\left[\frac{(\csc\theta-\cot\theta)+(\csc\theta+\cot\theta)}{(\csc\theta+\cot\theta)(\csc\theta-\cot\theta)}\right]=\sin\theta\left[\frac{2\csc\theta}{\csc^2\theta-\cot^2\theta}\right]$$

$$\text{LHS}=\sin\theta\left[\frac{2\csc\theta}{1}\right]=2\sin\theta\csc\theta=2=\text{RHS}$$

1.2.38 Change the following data into a "less than type" distribution and draw its ogive.

Class Interval	30–40	40–50	50–60	60–70	70–80	80–90	90–100
Frequency	7	5	8	10	6	6	8

Solution: To construct the "less than type" cumulative frequency distribution, take the upper class limits of each class interval and compute the cumulative frequencies by progressively summing the frequencies[cite: 3]:

Marks (Class Interval)	Frequency (f)	Cumulative Frequency (cf)
30–40	7	7
40–50	5	$7+5=12$
50–60	8	$12+8=20$
60–70	10	$20+10=30$
70–80	6	$30+6=36$
80–90	6	$36+6=42$
90–100	8	$42+8=50$
Total	$N=50$	

The points to be plotted on the Cartesian plane with the upper class limits on the horizontal axis (x-axis) and the cumulative frequencies on the vertical axis (y-axis) are:

$$(30,0),\ (40,7),\ (50,12),\ (60,20),\ (70,30),\ (80,36),\ (90,42),\ (100,50)$$

CURVE NEED TO BE ADDED BY PYTHON CODE

Determining Median from the Ogive:

Here, the total number of observations is $N=50$:

$$\frac{N}{2}=\frac{50}{2}=25$$

Locate 25 on the y-axis (cumulative frequency axis). From this point, draw a horizontal line parallel to the x-axis intersecting the ogive curve. From the intersection

point, drop a vertical perpendicular line to the x-axis.
The abscissa of this point on the x-axis gives the median:

$$\text{Median} = 65$$

1.2.39 Construct an equilateral triangle ABC with each side 5 cm. Then construct another triangle whose sides are $\frac{2}{3}$ times the corresponding sides of triangle ABC.
Solution: Set vertex B as the reference coordinate origin $\mathbf{0} \in \mathbb{R}^2$. The vertices of the master equilateral triangle are parameterized by:

$$\mathbf{v}_B = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_C = \begin{bmatrix} 5 \\ 0 \end{bmatrix}, \quad \mathbf{v}_A = \begin{bmatrix} 5\cos(60^\circ) \\ 5\sin(60^\circ) \end{bmatrix} = \begin{bmatrix} 2.5 \\ 2.5\sqrt{3} \end{bmatrix}$$

The geometric scaling corresponds to an isotropic dilation operator $\mathbf{H}_s = s\mathbf{I}_2$ with scale factor $s = \frac{2}{3}$:

$$\mathbf{H}_{2/3} = \begin{bmatrix} \frac{2}{3} & 0 \\ 0 & \frac{2}{3} \end{bmatrix}$$

Transforming the vertex coordinates:

$$\mathbf{v}_{C'} = \mathbf{H}_{2/3}\mathbf{v}_C = \begin{bmatrix} \frac{2}{3} & 0 \\ 0 & \frac{2}{3} \end{bmatrix} \begin{bmatrix} 5 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{10}{3} \\ 0 \end{bmatrix}$$

$$\mathbf{v}_{A'} = \mathbf{H}_{2/3}\mathbf{v}_A = \begin{bmatrix} \frac{2}{3} & 0 \\ 0 & \frac{2}{3} \end{bmatrix} \begin{bmatrix} 2.5 \\ 2.5\sqrt{3} \end{bmatrix} = \begin{bmatrix} \frac{5}{3} \\ \frac{5\sqrt{3}}{3} \end{bmatrix}$$

The side length of the scaled simplex is:

$$\|\mathbf{v}_{C'} - \mathbf{v}_B\|_2 = \frac{10}{3} \text{ cm} \approx 3.33 \text{ cm}$$

1.2.40 Draw two concentric circles of radii 2 cm and 5 cm. Take a point P on the outer circle and construct a pair of tangents PA and PB to the smaller circle. Measure PA.
IMAGE NEED TO BE ADDED.

1.2.41 A solid iron pole consists of a cylinder of height 220 cm and base diameter 24 cm, which is surmounted by another cylinder of height 60 cm and radius 8 cm. Find the mass of the pole, given that 1 cm^3 of iron has approximately 8 gm mass. (Use $\pi = 3.14$.)
Solution: The iron pole consists of two coaxial right circular cylinders stacked vertically[cite: 3].
Lower (Larger) Cylinder:

$$\text{Height, } h_1 = 220 \text{ cm}$$

$$\text{Base diameter, } d_1 = 24 \text{ cm} \implies \text{Radius, } r_1 = \frac{24}{2} = 12 \text{ cm}$$

The volume of the lower cylinder is:

$$V_1 = \pi r_1^2 h_1 = 3.14 \times (12)^2 \times 220 = 3.14 \times 31680 = 99475.2 \text{ cm}^3$$

Upper (Smaller) Cylinder:

$$\text{Height, } h_2 = 60 \text{ cm} \qquad \text{Radius, } r_2 = 8 \text{ cm}$$

The volume of the upper cylinder is:

$$V_2 = \pi r_2^2 h_2 = 3.14 \times 3840 = 12057.6 \text{ cm}^3$$

Total Volume of the Pole:

$$V = V_1 + V_2 = 99475.2 + 12057.6 = 111532.8 \text{ cm}^3$$

Mass of the Pole:
the mass of 1 cm^3 of iron is approximately 8 gm[cite: 3]:

$$\text{Total mass} = V \times 8 \text{ gm} = 111532.8 \times 8 = 892262.4 \text{ gm}$$

Converting the mass into kilograms:

$$\text{Mass in kg} = \frac{892262.4}{1000} = 892.2624 \text{ kg} \approx 892.26 \text{ kg}$$